

On-demand supply chain configuration and planning for shared additive manufacturing

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Abstract

Shared additive manufacturing (AM) platforms enable independent service providers to jointly fulfill customer orders by temporarily coordinating distributed resources. Unlike fixed supply chains, these platforms must configure supply chains at the order level, selecting providers and planning production across heterogeneous machines under cost and lead time constraints. This study addresses the supply chain configuration and planning (SCCP) problem in shared AM environments involving multiple providers with different capabilities. We propose an integrated optimization framework that combines provider selection with production planning through a hierarchical decision structure: provider selection, part-to-job grouping, and job-to-machine assignment with daily scheduling. The problem is formulated as a mixed-integer linear programming (MILP) model capturing machine heterogeneity, material compatibility, capacity constraints, and provider-specific operational policies. To improve scalability, a genetic algorithm (GA) with a three-layer chromosome is developed. Computational experiments show that GA achieves near-optimal solutions within 1% of MILP for medium-scale instances and remains effective for large-scale cases where MILP becomes intractable. Analyses further indicate that larger orders expand the provider network in a cost-driven manner, while tighter lead time requirements shift configurations toward premium providers and sharply increase total cost.

Keywords: additive manufacturing; shared additive manufacturing;
supply chain configuration and planning; on-demand manufacturing;
genetic algorithm

1. Introduction

Additive manufacturing (AM) has attracted increasing attention for its ability to support flexible and decentralized production and to enable on-demand manufacturing (Huang et al. 2013; Mellor, Hao, and Zhang 2014; Cuellar et al. 2018; Savolainen and Collan 2020). Ongoing advances in AM technologies, together with declining equipment costs and expanding material portfolios (Petch 2025; Nazir et al. 2023), have renewed interest in manufacturing paradigms that move beyond traditional systems centered on mass production (Kim and Lee 2023).

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AM technologies are characterized by layer-by-layer fabrication, enabling customized parts to be produced directly from digital designs without part-specific tooling (ISO/ASTM 2021; Hopkinson and Dickens 2003). Unlike conventional manufacturing processes that require substantial setup and tooling investment to achieve scale economies, AM involves minimal setup effort and allows rapid switching between products (Gibson, Rosen, and Stucker 2010; Cuellar et al. 2018). Although per-unit build times may be longer than those of high-volume processes, this flexibility makes AM particularly suitable for low-volume, high-variety production (Atzeni et al. 2010). Combined with fully digital workflows, these characteristics position AM as a natural enabler of on-demand, order-driven manufacturing (Gibson, Rosen, and Stucker 2010; Savolainen and Collan 2020).

In this context, shared manufacturing has emerged as a promising direction (Kim and Jung 2026), in which manufacturing resources owned by independent providers are temporarily connected to fulfill customer orders (Yu et al. 2020). Rather than relying on fixed production networks, shared manufacturing envisions dynamically configuring distributed resources on demand, leveraging excess capacity across multiple manufacturers (Wang et al. 2021). When combined with AM, this concept offers significant potential for delivering customized, one-of-a-kind products while lowering barriers to access for both customers and providers (Kim and Jung 2026).

In shared AM environments, customer orders often consist of multiple parts with heterogeneous material and geometric requirements, while individual providers possess limited capacity and specialized capabilities. Efficient order fulfillment therefore requires distributing production across multiple providers according to their available resources. However, existing research on shared AM has largely focused on high-level system architectures (Lupi et al. 2023) or matching orders to a single provider (Yang, Chen, and Kumara 2021; Kim and Jung 2026), leaving the coordination of multiple providers for a single order insufficiently addressed.

Addressing this challenge requires integrated supply chain configuration and planning (SCCP) at the order level. Such decisions include selecting appropriate providers, allocating parts to heterogeneous machines, and scheduling production to satisfy cost and lead time requirements. This order-level SCCP problem differs fundamentally from traditional AM supply chain studies, which typically focus on long-term network design, such as facility location and capacity investment (Barz, Buer, and Haasis 2016; Jimo et al. 2022). In contrast, shared AM platforms must dynamically form temporary supply chains for each incoming order, tightly coupling provider selection with detailed production scheduling decisions.

However, existing literature does not systematically address this coupled, order-level SCCP problem in shared manufacturing contexts. While related studies on AM supply chain have examined provider selection (Akmal et al. 2022), AM supply chain network design (Barz, Buer, and Haasis 2016; Jimo et al. 2022), and production scheduling in AM (Kucukkoc 2019; Xueping Li and Zhang 2018), none simultaneously addresses the integrated configuration and detailed scheduling problem at the order level in shared AM environments.

To address these gaps, this study makes three main contributions. First, it proposes an on-demand SCCP framework for shared AM environments, where supply chains are dynamically configured for individual customer orders rather than designed for long-term operation.

Second, it develops an integrated optimization model that couples provider selection with detailed production planning, formulated as a mixed-integer linear programming (MILP) problem that explicitly accounts for machine heterogeneity, material compatibility, capacity constraints, and provider-specific operational policies. Third, it develops a scalable genetic algorithm (GA)-based solution approach to enable systematic analysis of how order characteristics shape configuration and planning decisions in shared AM settings.

The remainder of this paper is organized as follows. Section 2 reviews the related literature and identifies research gaps. Section 3 describes the shared AM environment and defines the on-demand SCCP problem. Section 4 presents the mathematical formulation as a MILP model. Section 5 develops a GA-based solution approach for scalable problem-solving. Section 6 evaluates the proposed solution approach and analyzes the impact of order characteristics on SCCP decisions through computational experiments. Finally, Section 7 concludes the paper with a summary of key findings, managerial insights, limitations, and directions for future research.

2. Literature review and research gaps

2.1. Literature review

AM has been increasingly recognized as a technology with the potential to reshape supply chain structures due to its flexibility, digital workflows, and reduced setup requirements. Although AM adoption as a primary production method remains limited, research at the intersection of AM and supply chain management has grown steadily (Kunovjanek, Knofius, and Reiner 2022).

Early studies emphasized that the fundamental characteristics of AM differ from those of conventional manufacturing and may alter optimal supply chain configurations (Walter, Holmström, and Yrjölä 2004; Barz, Buer, and Haasis 2016; Tziantopoulos et al. 2019). Building on this insight, several studies compared centralized and decentralized AM-based supply chain structures, particularly in the context of spare parts logistics.

Khajavi, Partanen, and Holmström (2014) analyzed aircraft spare parts supply chains under centralized and decentralized AM configurations and showed that centralized production is currently more cost-effective, while decentralized configurations may become viable as AM technologies mature. Similarly, Liu et al. (2014) used the SCOR model to compare conventional and AM-enabled spare parts supply chains, demonstrating that AM adoption reduces inventory levels and that centralized AM configurations perform better under low and volatile demand, whereas distributed configurations are preferable under high or stable demand.

Extending these comparisons, Li et al. (2019) showed that neither centralized nor distributed AM configurations consistently dominate in terms of cost or response time; instead, the optimal structure depends on demand characteristics and waiting penalty costs. To bridge the gap between these extremes, Khajavi, Holmström, and Partanen (2018) proposed an AM hub configuration that combines centralized efficiency with decentralized flexibility, and Xu, Rodgers, and Guo (2021) demonstrated its effectiveness in disruption-prone and time-critical environments.

Moving beyond predefined configuration comparisons, Chowdhury et al. (2019) developed a decision framework for designing AM-based supply chain networks. Their two-stage model jointly determines facility locations, capacities, and operational decisions, providing one of the earliest mathematical formulations for supply chains composed entirely of AM technologies.

While the studies above primarily address long-term supply chain configuration decisions, a separate stream of research has examined short-term operational decisions in AM-enabled environments. Hedenstierna et al. (2019) proposed a bidirectional partial outsourcing (BPO) scheme in which AM providers dynamically share capacity within a pre-existing partner network, demonstrating that collaborative capacity sharing improves cost efficiency and delivery performance. Building on the build-to-model concept, Akmal et al. (2022) developed a dynamic supplier selection practice in which the most suitable AM supplier is identified for each individual spare part, enabling on-demand procurement from multiple AM suppliers without substantial transaction costs. These studies effectively capture the potential of AM to shift supply chain decisions from long-term strategic commitments toward shorter-term operational flexibility.

Beyond the supply chain management perspective, the operational planning of AM has also been studied within the broader context of distributed manufacturing systems, where production is allocated across geographically dispersed sites. Zehetner and Gansterer (2022) introduced a collaborative multi-site batching problem in which a central authority batches and schedules orders across dispersed AM sites, and Kucukkoc (2024) developed a MILP model for multi-site AM machine scheduling that jointly minimizes production and transportation costs together with carbon emissions. While these studies capture the distributed nature of AM production, they schedule parts within a fixed set of sites and machines defined prior to optimization, and typically assume a central coordinating authority. The formation of the network itself, including the selection of independent providers, is not treated as a decision.

Table 1 summarizes the key studies, comparing their problem type, decision horizon, network type, and methodology. As shown in the table, existing research either addresses long-term configuration decisions within integrated supply chains or short-term decisions that are limited in scope, such as capacity sharing within pre-established networks, part-by-part supplier selection, or machine scheduling within a predefined set of production sites. In contrast, this study addresses the integrated, order-level configuration of an on-demand supply chain network for multi-part orders across independent AM service providers.

2.2. Research gaps

While supply chain studies that incorporate AM have been steadily expanding, several key gaps remain. This section outlines limitations in the existing literature and explains how this study aims to address them.

First, most prior studies focus on long-term supply chain configuration decisions, such as facility location and capacity planning, based on demand forecasts. However, supply chains in shared AM environments are rather not fixed over time and are instead dynamically formed and dissolved based on individual orders. While some recent studies have explored short-term

Table 1. Comparison of existing studies on supply chains configuration and planning.

Study	Problem type	Decision horizon	Network type	Methodology
Khajavi et al. (2014)	Configuration comparison	Long-term	Integrated supply chain network	Analytical
Liu et al. (2014)	Configuration comparison	Long-term	Integrated supply chain network	Analytical
Khajavi et al. (2018)	Configuration comparison	Long-term	Integrated supply chain network	Simulation
Li et al. (2019)	Configuration comparison	Long-term	Integrated supply chain network	Simulation
Chowdhury et al. (2019)	Network design	Long-term	Integrated supply chain network (AM only)	Optimization
Xu et al. (2021)	Configuration comparison	Long-term	Integrated supply chain network	Simulation
Hedenstierna et al. (2019)	Capacity sharing	Short-term	Pre-existing partner network	Analytical
Akmal et al. (2022)	Supplier selection	Short-term (part-level)	Independent AM suppliers	Design science
Zehetner and Gansterer (2022)	Batching & scheduling	Short-term	Predefined distributed sites	Optimization
Kucukkoc (2024)	Machine scheduling	Short-term	Predefined distributed sites	Optimization
The current paper	Provider selection & planning	Short-term (order-level)	Independent AM service provider marketplace	Optimization

operational decisions in AM environments (Hedenstierna et al., 2019; Akmal et al., 2022; Zehetner and Gansterer, 2022; Kucukkoc, 2024), these approaches either assume pre-established partner networks, schedule production across a predefined set of distributed sites, or address supplier selection on a part-by-part basis. Accordingly, there is a need for an integrated framework that configures on-demand supply chain networks at the order level in real time, dynamically assigning the multi-part order to AM service providers with heterogeneous capabilities.

Second, most studies assume limited product variety and oversimplified AM machine characteristics in the context of production management. However, in practice, shared AM platforms must accommodate diverse part types and materials across machines with varying technologies, such as fused deposition modeling (FDM), stereolithography (SLA), and selective laser sintering (SLS), as well as different operational capabilities. This necessitates integrating operational-level decisions (e.g., machine assignment and production scheduling) into provider selection, which existing configuration models typically overlook.

Finally, existing literature often centers on spare parts supply chains, particularly in the aviation sector. However, the shared AM setting and the operational complexity explored in this study are applicable to a broader range of customization-driven industries. This highlights the need for generalizable models that support more diverse use cases.

To address these gaps, this study proposes an integrated SCCP framework for shared AM environments. Our approach combines AM service provider selection with detailed production planning decisions, capturing the interdependencies between configuration choices and operational constraints. By formulating this as an optimization problem and developing both exact and heuristic solution methods, we provide a systematic approach to automate supply chain formation in response to dynamic order characteristics.

3. Problem description

The emergence of shared AM platforms has enabled new forms of supply chain coordination, where production networks are assembled dynamically for each customer order rather than established as fixed long-term relationships. In shared AM environments, multiple independent service providers with heterogeneous AM machines offer their resources on-demand, and customers can leverage this distributed capacity to fulfill their production needs. Customers can determine a supply flow tailored to their order by selecting the best combination of part manufacturers, a focal provider responsible for assembly and delivery, and the associated logistics flows, according to their own priorities such as cost or lead time.

We refer to such supply chains as *on-demand supply chains*. These on-demand supply chains differ fundamentally from traditional company-driven supply chains. While traditional supply chains, which are typically static and optimized for mass production, on-demand supply chains are configured dynamically to respond to individual customer needs. Once an order is fulfilled, the supply chain dissolves. Since these supply chains are order-specific, multiple configurations can exist in parallel; a single service provider may simultaneously participate in different roles across multiple active supply chains.

As the on-demand supply chain in shared AM is configured on a per-order basis, its structure is closely tied to the underlying production planning decisions for the AM machines involved (such as part-to-job grouping and job scheduling). In other words, determining which machines from which providers are used to fulfill a given order is an integral part of the configuration itself.

For example, in Figure 1, the supply chain for Order 1 (left) consists of Providers A, B, and D. This configuration was selected not arbitrarily but because the production plan that utilizes one machine from Provider A, two machines from Provider B, and two machines from Provider D proved to be the most efficient. The selection of these machines is in turn

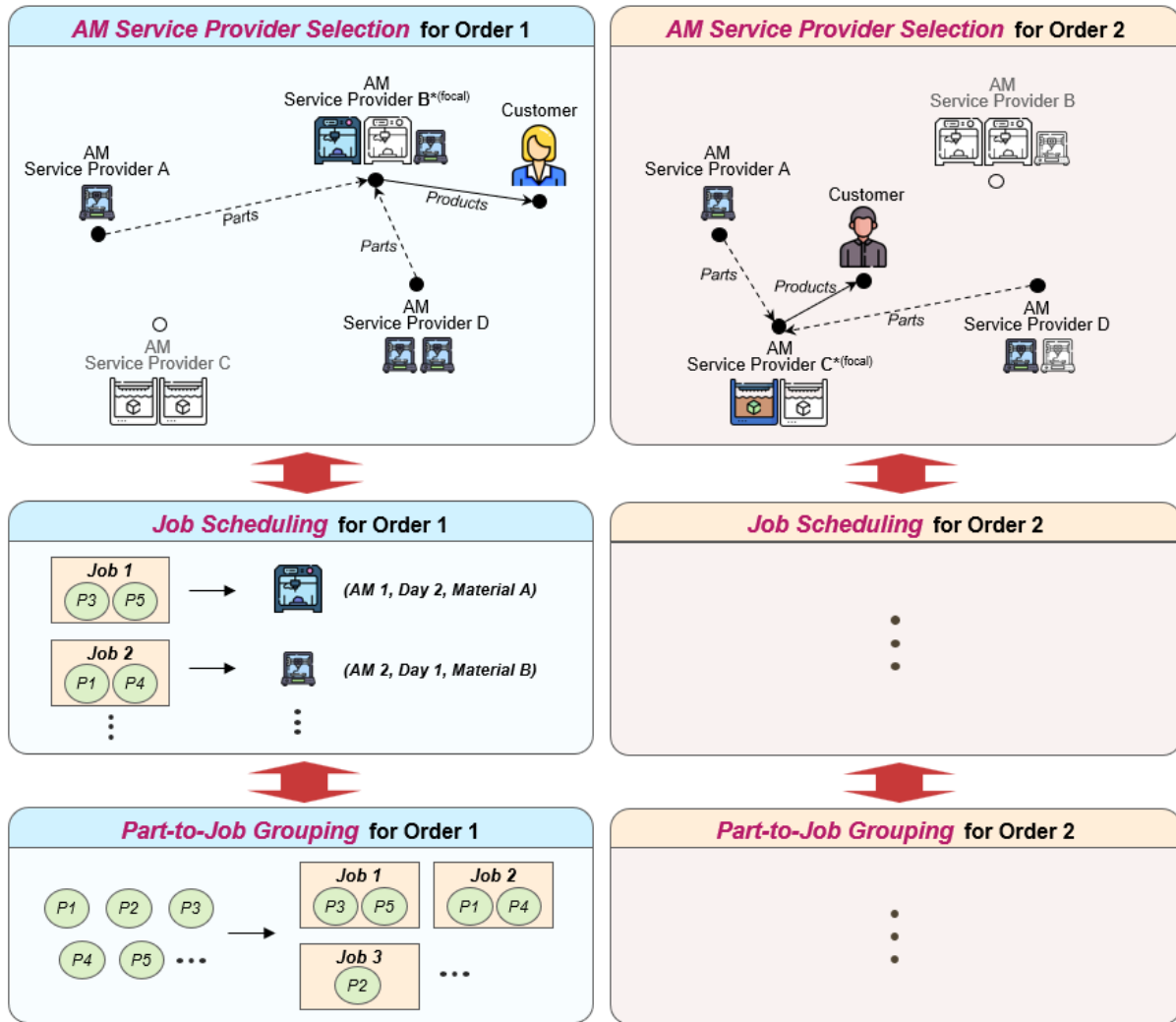


Figure 1. On-demand supply chain configurations and the three-level hierarchical decision structure.

shaped by the underlying part-to-job grouping and job scheduling decisions, as illustrated in the lower portion of Figure 1. Similarly, for Order 2 (right), the most efficient production plan required only one machine each from Providers A, C, and D. As a result, the supply chain for Order 2 was configured accordingly to include these three providers.

In this study, we term this integrated decision-making process *on-demand supply chain configuration and planning (SCCP)*. Configuration determines the network structure: (1) which service providers participate in fulfilling the order, and (2) which service provider serves as the focal provider for assembly and delivery. Planning determines the operational details: (3) how production tasks (parts and jobs) are allocated among selected providers and their machines, and (4) when each job is scheduled for production to ensure coordinated completion and timely delivery.

To summarize, the SCCP problem involves a three-level hierarchical decision structure, where each level is interdependent and influences the others.

At the network level, the problem determines which AM service providers participate in fulfilling the order and designates one as the focal provider. This selection considers multiple

factors similar to supplier selection in traditional supply chains: machine capabilities (types, materials, capacity), cost structure (ordering, processing, coordination), operational patterns (schedules, availability), and spatial proximity (to other providers and customers). The outcome of this level defines the available resource pool for production planning, while conversely, the machine and scheduling requirements identified at the other two levels may also influence which providers are ultimately selected.

At the machine level, jobs are allocated to specific machines and scheduled on particular dates. This allocation must respect machine availability, capacity limits, and material compatibility while ensuring coordinated completion across providers to enable timely assembly and delivery. The machine-level decisions are shaped by the provider configuration established at the network level, as the set of available machines depends on which providers are selected. At the same time, the grouping structure determined at the job level defines the production requirements that machine assignments must accommodate.

At the job level, parts are grouped into jobs based on their characteristics and compatibility requirements. This grouping must balance production efficiency with physical and operational constraints. Job-level decisions are directly shaped by machine-level assignments, as the feasibility of a grouping depends on which machines are available for allocation. The provider configuration at the network level also plays a role indirectly, as it determines the set of machines from which assignments can be made. Conversely, the resulting grouping structure defines the production requirements that machine-level assignments must accommodate, and may also influence which providers are ultimately needed at the network level.

The three-level hierarchical structure described above reflects a fundamentally different approach compared to traditional supply chain design, as summarized in Table 2, across multiple dimensions. In terms of planning horizon and network structure, traditional supply chains are established as long-term, fixed structures optimized for multi-order operation under demand forecasts (Vidal and Goetschalckx 1997; Melo, Nickel, and Saldanha-da-Gama 2009), with suppliers participating through exclusive or long-term contracts (Simchi-Levi, Kaminsky, and Simchi-Levi 2004). In contrast, on-demand supply chains are triggered by individual customer orders rather than company-driven strategic planning. The supply network is assembled dynamically according to the characteristics of each order, such as the number of parts and manufacturing requirements, as well as customer priorities such as cost or lead time, and dissolves upon order completion.

In terms of operational decisions, traditional supply chain design typically focuses on network-level decisions such as facility location and aggregate capacity planning (Vidal and Goetschalckx 1997; Melo, Nickel, and Saldanha-da-Gama 2009), rarely extending to short-term, order-level scheduling decisions. As a result, decision-making typically proceeds hierarchically and sequentially, from network configuration to production planning and short-term operations. In contrast, on-demand SCCP requires an integrated optimization spanning provider selection, job grouping, and machine-level scheduling, where these decision layers are tightly interdependent.

Table 2. Comparison between traditional supply chain design and on-demand SCCP.

Aspect	Traditional Supply Chain Design	On-Demand SCCP in Shared AM
Planning horizon	Long-term, strategic	Order-specific, dynamic
Network structure	Established in advance, multi-order operation	Configured on-demand, dissolves after order
Decision maker	Company-driven	Customer-driven
Production strategy	Mass production, economies of scale	Customized production, on-demand capacity
Resource allocation	Aggregate capacity planning	Machine-level, part-level assignment
Supplier participation	Exclusive or long-term contracts	Simultaneous participation in multiple configurations
Optimization focus	Network design (location, capacity)	Integrated configuration and detailed production planning
Decision layers	Typically hierarchical, sequential	Multi-level, tightly interdependent

4. Mathematical model

4.1. Model formulation

Figure 2 illustrates the decision structure of the on-demand SCCP problem in shared AM environments. The model encompasses distributed parts that must be grouped into jobs, allocated to AM machines operated by selected service providers, and scheduled within provider-specific operating windows. After manufacturing, all parts are shipped to a designated focal provider for final assembly and delivery to the customer. Note that all decisions are optimized simultaneously, with part grouping, machine assignment, provider selection, and scheduling mutually dependent on one another.

The problem is formulated as a MILP model minimizing total cost, with outputs including the supply chain configuration (selected providers and focal provider designation), production planning decisions (part-to-job grouping, machine assignment, and material selection), and daily job allocation. The model is developed under the following assumptions.

- Provider operating hours, schedules, and machine specifications are deterministic.
- Setup time is incurred at the start of each production operation. Only parts made of identical materials can be grouped and printed together as a single job.
- Post-processing time is constant across providers. Material changeover time between jobs is not considered.
- Focal provider assembly operates continuously (24-hour basis), independent of machine schedules.

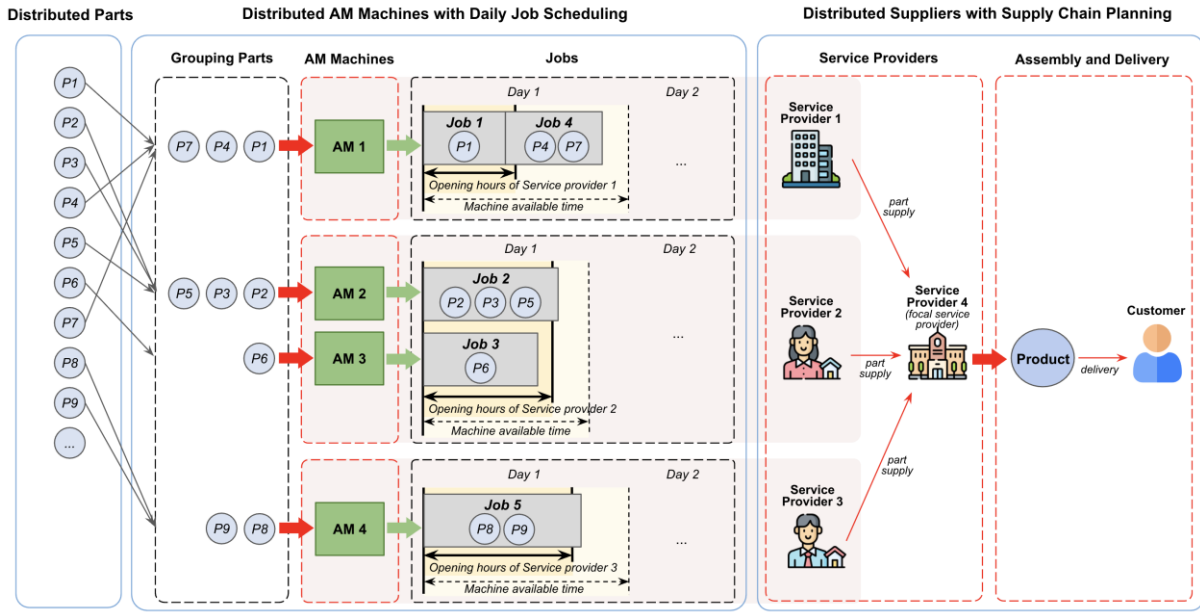


Figure 2. Concept model.

- Completed parts are delivered together; partial deliveries are not permitted.
- Machine capacities and build volumes are fixed with no breakdowns during the planning horizon.
- Transportation times and costs are deterministic and known.

sRelated sets, parameters, and decision variables are presented in Appendix A.

The objective function of this MILP model is given in Equation (1), minimizing the total production and delivery costs. Provider costs reflect pricing policies on machine usage, material handling, and focal service. Bilinear terms (e.g., $Z_k^{SP} Z_{k'}^{foc}$) are linearized using auxiliary variables and additional constraints.

$$\text{minimize } TC = \sum_k C_k^{SP} + c^{\text{del}} \left(\sum_{k,k',k \neq k'} d_{k,k'} Z_k^{SP} Z_{k'}^{foc} + \sum_k d_k^{\text{cust}} Z_k^{foc} \right) \quad (1)$$

Part assignment and job-machine mapping

$$\sum_j X_{ij} = 1 \quad \forall i \quad (2)$$

$$Y_{jmt} \sum_i b_i X_{ij} \leq \beta_m \quad \forall j, m, t \quad (3)$$

$$\sum_{m,t} Y_{jmt} \leq 1 \quad \forall j \quad (4)$$

$$|I| \times \sum_{m,t} Y_{jmt} \geq \sum_i X_{ij} \quad \forall j \quad (5)$$

$$\sum_{m,t} Y_{jmt} \leq \sum_i X_{ij} \quad \forall j \quad (6)$$

Equations (2)–(6) define the part assignment and job-machine relationships. Specifically, Equation (2) ensures that each part is assigned to exactly one job, while Equation (3) limits the total area of parts not exceeding machine bed capacity (Li et al. 2017). Equation (4) prevents a job from being split across multiple machines or time slots. Lastly, Equations (5)–(6) link part assignments to job scheduling, ensuring jobs are assigned to specific machines and days.

Machine usage and availability

$$P_{jmt} = Y_{jmt} \left(s_m + \delta_m \sum_i v_i X_{ij} \right) \quad \forall j, m, t \quad (7)$$

$$T_{mt} = \sum_j P_{jmt} + u_{mk} \quad \forall m, t \quad (8)$$

$$T_{mt} \leq h_{mt} \quad \forall m, t \quad (9)$$

$$T_{mt}^{\text{long}} = \max_j \{P_{jmt}\} \quad \forall m, t \quad (10)$$

$$u_{mk} (T_{mt} - T_{mt}^{\text{long}}) \leq o_{kt} \quad \forall k, m, t \quad (11)$$

Equations (7)–(11) govern machine usage and provider availability. Equation (7) computes job processing time, and Equation (8) aggregates daily machine usage.

Equation (9) prevents total machine usage from exceeding its available time. To prevent job initiation outside of operating hours, Equation (10) identifies the longest fabrication time among all jobs per machine-day. Based on this, Equation (11) restricts total processing time minus longest job time to within available hours (Kim and Jung 2026).

Cost computation

$$T_{mt}^{\text{hour}} \geq T_{mt} \quad \forall m, t \quad (12)$$

$$T_{mt}^{\text{hour}} \leq T_{mt} + K_{mt}^{\text{AM}} \quad \forall m, t \quad (13)$$

$$T_{mt}^{\text{extra}} \geq T_{mt}^{\text{hour}} - \tau_m^{\text{AM}} \quad \forall m, t \quad (14)$$

$$T_{mt}^{\text{extra}} \leq h_{mt} K_{mt}^{\text{AM}} \quad \forall m, t \quad (15)$$

$$C_{mt}^{\text{AM}} = c_m^{\text{basic}} K_{mt}^{\text{AM}} + c_m^{\text{extra}} T_{mt}^{\text{extra}} \quad \forall m, t \quad (16)$$

$$C_k^{\text{SP}} = c_k^{\text{order}} Z_k^{\text{SP}} + \sum_{m,t} u_{mk} C_{mt}^{\text{AM}} + c_k^{\text{rm}} (N_k^{\text{rm}} - 1) + c_k^{\text{foc}} Z_k^{\text{foc}} \quad \forall k \quad (17)$$

Equations (12)–(15) define time-based variables used for machine usage fees. While Kim and Jung (2025) estimated the fee using two separate expressions, this study introduces an explicit variable T_{mt}^{extra} for excess usage time computed via Equations (14)–(15), and applied in

the fee calculation (Equation (16)) for more accurate cost estimation. Equation (17) calculates total provider cost C_k^{SP} .

Linking production and operation variables

$$K_{kt}^{SP} \geq u_{mk} Y_{jmt} \quad \forall j, k, m, t \quad (18)$$

$$K_{kt}^{SP} \leq \sum_{j,m} u_{mk} Y_{jmt} \quad \forall k, t \quad (19)$$

$$K_{mt}^{AM} \geq \sum_j \frac{Y_{jmt}}{|J|} \quad \forall m, t \quad (20)$$

$$K_{mt}^{AM} \leq \sum_j Y_{jmt} \quad \forall m, t \quad (21)$$

Equations (18)–(21) link job assignments to provider and machine activation indicators, K_{kt}^{SP} and K_{mt}^{AM} .

Raw material selection and availability

$$R_{jmtr} \leq \gamma_{mr} \quad \forall j, m, t, r \quad (22)$$

$$R_{jmtr} \leq \alpha_{ir} + (1 - X_{ij}) \quad \forall i, j, m, t, r \quad (23)$$

$$\sum_r R_{jmtr} = Y_{jmt} \quad \forall j, m, t \quad (24)$$

$$\sum_{j,t} R_{jmtr} \leq |J| \times |T| \times R_{kmr}^{aux} \quad \forall k, m, r \text{ such that } u_{mk} = 1 \quad (25)$$

$$\sum_{j,t} R_{jmtr} \geq R_{kmr}^{aux} \quad \forall k, m, r \text{ such that } u_{mk} = 1 \quad (26)$$

$$\Omega_{kr} \geq R_{kmr}^{aux} \quad \forall k, m, r \text{ such that } u_{mk} = 1 \quad (27)$$

$$\Omega_{kr} \leq \sum_m u_{mk} R_{kmr}^{aux} \quad \forall k, r \quad (28)$$

$$N_k^{rm} = \sum_r \Omega_{kr} \quad \forall k \quad (29)$$

Equations (22)–(24) ensure raw material assignments respect machine capability constraints. To determine whether material type r is used by provider k , the binary variable Ω_{kr} is computed via Equations (25)–(28) using auxiliary variable R_{kmr}^{aux} . While Kim and Jung (2025) required one constraint per provider, the proposed model uses four constraints to track material usage across multiple providers. Accordingly, the number of material types per provider is calculated in Equation (29).

Part production completion time

$$S_{kt} \geq K_{kt}^{SP} - \sum_{t+1}^T K_{kt}^{SP} \quad \forall k, t \quad (30)$$

$$S_{kt} \leq Z_k^{\text{SP}} \quad \forall k, t \quad (31)$$

$$\sum_t S_{kt} = Z_k^{\text{SP}} \quad \forall k \quad (32)$$

$$T_{kt}^{\text{long_day}} = \max_m \{u_{mk} T_{mt}\} \quad \forall k, t \quad (33)$$

$$\text{big } M \times D_{kt}^{\text{extra_aux}} \geq T_{kt}^{\text{long_day}} - o_{kt} + t^{\text{post}} \quad \forall k, t \quad (34)$$

$$D_{kt}^{\text{extra}} = S_{kt} D_{kt}^{\text{extra_aux}} \quad \forall k, t \quad (35)$$

$$PCD_k = \sum_t \{t \times S_{kt} + d_{kt}^{\text{next}} D_{kt}^{\text{extra}}\} \quad \forall k \quad (36)$$

$$T_k^{\text{Part}} = (PCD_k - Z_k^{\text{SP}}) + \sum_t \left\{ \frac{(S_{kt} - D_{kt}^{\text{extra}}) T_{kt}^{\text{long_day}}}{24} \right\} + \frac{t^{\text{post}} Z_k^{\text{SP}}}{24} \quad \forall k \quad (37)$$

Equations (30)–(36), adopted and extended from Kim and Jung (2025), define the final job scheduling and compute integer-valued part completion days. Variable S_{kt} indicates the starting day of the final job at provider k (renamed from MS_t for clarity), and is defined only when the provider is selected. The bilinear term in Equation (35) is linearized using standard techniques.

While the previous study uses integer completion days, the proposed model introduces continuous completion time (Equation (37)) to enable more accurate modeling of delivery and final assembly. This combines: (1) working days before completion, (2) intra-day production hours on the final day, and (3) post-processing time.

Supply chain configuration

$$\sum_k Z_k^{\text{SP}} \geq 1 \quad (38)$$

$$Z_k^{\text{SP}} \leq \sum_t K_{kt}^{\text{SP}} \quad \forall k \quad (39)$$

$$\sum_k \sigma_k Z_k^{\text{SP}} \geq \psi \sum_k Z_k^{\text{SP}} \quad (40)$$

$$\sum_k Z_k^{\text{foc}} = 1 \quad (41)$$

$$Z_k^{\text{foc}} \leq \kappa_k Z_k^{\text{SP}} \quad \forall k \quad (42)$$

Equations (38)–(42) define the selection of AM service providers and the designation of the focal provider in the supply chain. Equation (38) ensures that at least one provider is selected to fulfill the order. Equation (40) requires that the average rating score of the selected providers exceed the threshold value (ϕ), where σ_k denotes the rating score of service provider k . In line with the sharing economy environment (Yang, Chen, and Kumara 2021), this score reflects past performance and may include customer evaluations from previous orders as well as ratings from other providers who have collaborated with the provider.

Equation (41) ensures that exactly one focal provider is selected, and this provider must be among the candidates allowed by the binary parameter κ_k in Equation (42). The binary parameter κ_k is set to 1 when provider k is willing and able to serve as the focal provider of the supply chain, performing the subsequent stages such as assembly, product finalization, and transfer for delivery, for which it receives additional compensation c_k^{foc} .

Delivery, assembly, and finalization

$$T_k^{\text{Arrival}} \geq T_k^{\text{Part}} + \sum_{k'=1}^K \left(\frac{d_{k,k'}}{v} Z_{k'}^{\text{foc}} \right) - \text{big M} \times (1 - Z_k^{\text{SP}}) \quad \forall k \quad (43)$$

$$T_k^{\text{Arrival}} \leq \text{big M} \times Z_k^{\text{SP}} \quad \forall k \quad (44)$$

$$T^{\text{final}} = \max_k \{T_k^{\text{Arrival}}\} + \frac{t^{\text{asby}}}{24} \quad (45)$$

$$LT = T^{\text{final}} + \sum_k \frac{d_k^{\text{cust}}}{v} Z_k^{\text{foc}} \quad (46)$$

After part manufacturing, the fabricated parts are sent to the focal provider for assembly into the final products. Equation (43) calculates the arrival time of parts delivered from provider k to the focal provider, as the sum of the part fabrication time at provider k and the delivery time to the focal provider. Equation (45) computes the completion time of the order at the focal provider, including the assembly and product finalization time (t^{asby}). On this value, adding the delivery time to the customer yields the lead time LT in Equation (46).

Planning horizon and start condition

$$\sum_{j,m} Y_{jm1} \geq 1 \quad (47)$$

$$LT \leq |T| \quad (48)$$

Equations (47)–(48) ensure that production starts on day one and is completed within the customer-specified planning horizon.

Variable constraints

$$\begin{aligned} X_{ij}, Y_{jmt}, R_{jmtr}, Z_k^{\text{foc}}, Z_k^{\text{SP}}, S_{kt}, K_{mt}^{\text{AM}}, K_{kt}^{\text{SP}}, \Omega_{kr}, R_{kmr}^{\text{aux}}, D_{kt}^{\text{extra}}, D_{kt}^{\text{extra_aux}} \in \{0, 1\}, \\ P_{jmt}, T_{mt}, T_{mt}^{\text{long}}, T_{kt}^{\text{long_day}}, T_k^{\text{part}}, T_k^{\text{Arrival}}, T_k^{\text{Final}}, C_{mt}^{\text{AM}}, C_k^{\text{SP}}, LT \geq 0, \\ T_{mt}^{\text{hour}}, T_{mt}^{\text{extra}}, N_k^{\text{mat}}, PCD_k \in \mathbb{Z}_+ \quad \forall i, j, k, m, t, r \end{aligned} \quad (49)$$

Lastly, Equation (49) defines basic domain constraints on the variables.

Figure 3 illustrates the variable relationships and computation flow of the proposed model. The five key decision variables are simultaneously optimized, with derived variables feeding into the objective function value and lead time.

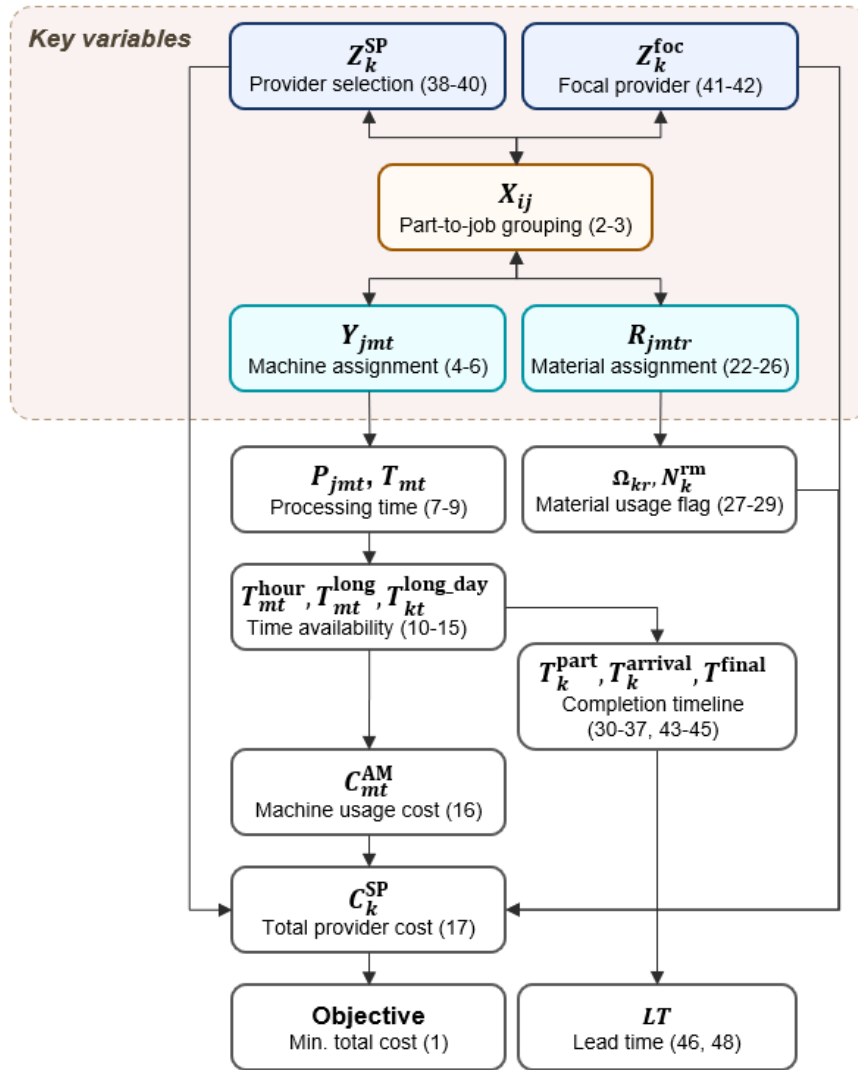


Figure 3. Variable relationships and equation numbers in the proposed MILP model.

4.2. Basic application of the MILP model

To illustrate the impact of extending the model to incorporate supply chain configuration, the dataset from Kim and Jung (2025) were reused: three providers with nine machines and an order requiring 33 parts (three sets of 11 parts). Provider schedules, machine specifications, and part requirements follow the original study. Additional parameters introduced for the extended supply chain context include: focal provider incentive (5,000 KRW), inter-provider distances (17 km), customer distance from each provider (10 km), delivery speed (30 km/h), unit delivery cost (200 KRW/km), and final assembly time at focal provider (3 hours).

While the original study used a bi-objective model, the present study considers a single-objective approach minimizing total cost. Accordingly, a single experimental setting is applied in the basic application, and the resulting compared against the cost-only scenario reported in the original study. Experiments were conducted using Python 3.11.5 and Gurobi 11.0.0 on an Intel Core i7-10700 CPU, with time limit of 600 seconds (to match the original study), and 3,600 seconds.

Table 3 compares results between the single-provider model (Kim and Jung 2026) and the proposed multi-provider model. Total cost increases from 39,500 KRW to 47,000 KRW (600s) and 46,500 KRW (3,600s), primarily due to new cost components: focal provider coordination (5,000 KRW) and delivery (2,000 KRW).

Table 3. Cost comparison with the original study (Kim and Jung 2026).

	Single provider model (original study)	Multi provider model (this study, 600s)	Multi provider model (this study, 3,600s)
Total cost (KRW)	39,500	47,000	46,500
– Basic order cost	5,000	5,000	5,000
– Part production cost	34,500	35,000	34,500
– Subtotal (comparable)	39,500	40,000	39,500
– Focal provider cost	-	5,000	5,000
– Delivery cost	-	2,000	2,000

Notably, when only comparable cost components are considered (basic order cost and part production cost), the multi-provider model achieves 40,000 KRW within a 600-second time limit, which is little higher than the 39,500 KRW reported in the original study under the same limit. With an extended computation time of 3,600 seconds, the multi-provider model attains the same production cost of 39,500 KRW, matching the original study's production efficiency. This demonstrates that extending to multi-provider settings preserves production efficiency while enabling supply chain coordination.

However, the original study solved three separate models (one per provider) and manually selected the best result (Provider 3). In contrast, the proposed model simultaneously evaluates all providers within a unified framework and automatically determines optimal allocation. Although this increases computational complexity, the integrated formulation enables supply chain coordination decisions not supported in the original single-provider setting.

5. GA-based solution approach

The integrated SCCP problem is computationally challenging due to its hierarchical structure and strong interdependencies among three tightly coupled decisions: (1) provider selection including focal provider designation, (2) part-to-job grouping subject to material and capacity constraints, and (3) job assignment and daily scheduling on heterogeneous machines.

The problem exhibits computational complexity at multiple levels. Job scheduling on heterogeneous parallel machines with material compatibility and daily capacity constraints corresponds to an unrelated parallel machine scheduling problem with batch processing, which is strongly NP-hard (Kucukkoc 2019). Part-to-job grouping generalizes the NP-hard two-dimensional bin packing problem (Li and Zhang 2018), with the number of feasible groupings growing exponentially with part count. Provider selection further expands the

solution space by changing available machines, capabilities, and schedules, fundamentally reshaping the feasible region.

As demonstrated in Section 4.2, small instances (33 parts, three providers) required up to 3,600 seconds for the MILP solver to reach optimality. Preliminary experiments with larger instances show computation time increases rapidly with problem size, rendering exact methods impractical for realistically scaled problems.

Given these challenges, this study adopts a GA for its effectiveness in multi-layer combinatorial optimization in manufacturing and scheduling contexts (Zhang et al. 2022; Wen et al. 2023).

Figure 4 illustrates the GA procedure, in which an initial population is generated using the multilayer chromosome structure (Section 5.1) and evolves through selection, crossover, mutation, and fitness evaluation. Genetic operations are applied separately to the AM Service Provider Layer (SPL), Part-Job Layer (PJL), and Job Scheduling Layer (JL) to preserve the hierarchical decision structure while maintaining interdependencies among decisions (Zhang et al. 2022; Wen et al. 2023). The algorithm terminates when a predefined stopping criterion is met.

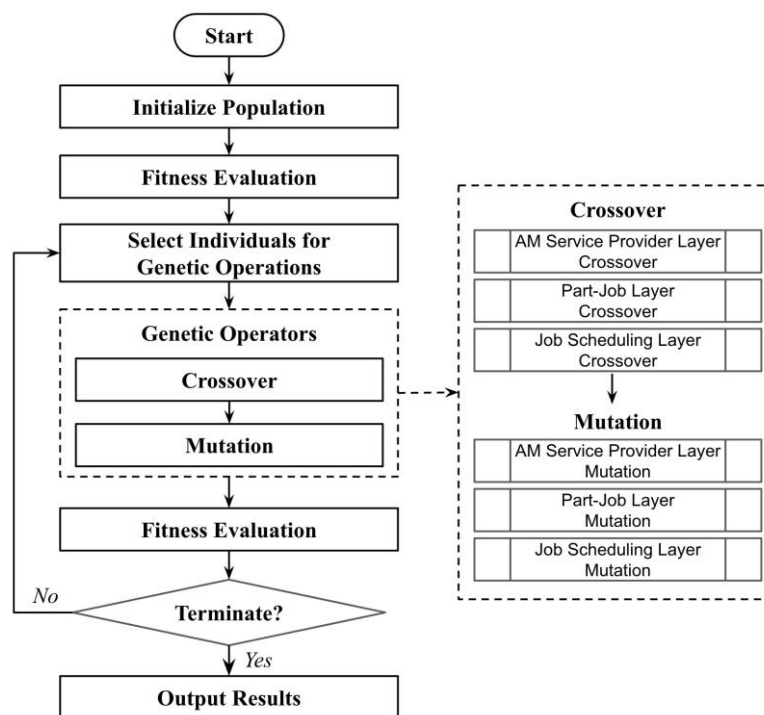


Figure 4. Overall GA flow.

5.1. Encoding

To represent the integrated SCCP problem, decision variables are encoded into a three-layer chromosome structure: (1) AM service provider selection, (2) part-to-job grouping, and (3) job scheduling. Such multi-layer representations are widely adopted for hierarchical combinatorial problems to capture interdependent decision effects (Zhang et al. 2022; Wen et al. 2023).

The SPL encodes provider participation using binary genes, with an additional gene indicating the focal provider. The PJL assigns each part to a job via gene values representing job indices. The JL encodes indices of pre-validated scheduling combinations, each corresponding to a feasible machine, production day, and material assignment. These indices are retrieved from a lookup table based on machine availability and material compatibility, reducing chromosome length and preventing infeasible assignments.

As an illustrative example, a sample input chromosome is shown in Figure 5. In the JL, each gene stores an index referencing a pre-validated scheduling option from the lookup table, which is decoded into the associated machine, production day, and raw material. This ensures that only feasible scheduling combinations are considered during decoding.

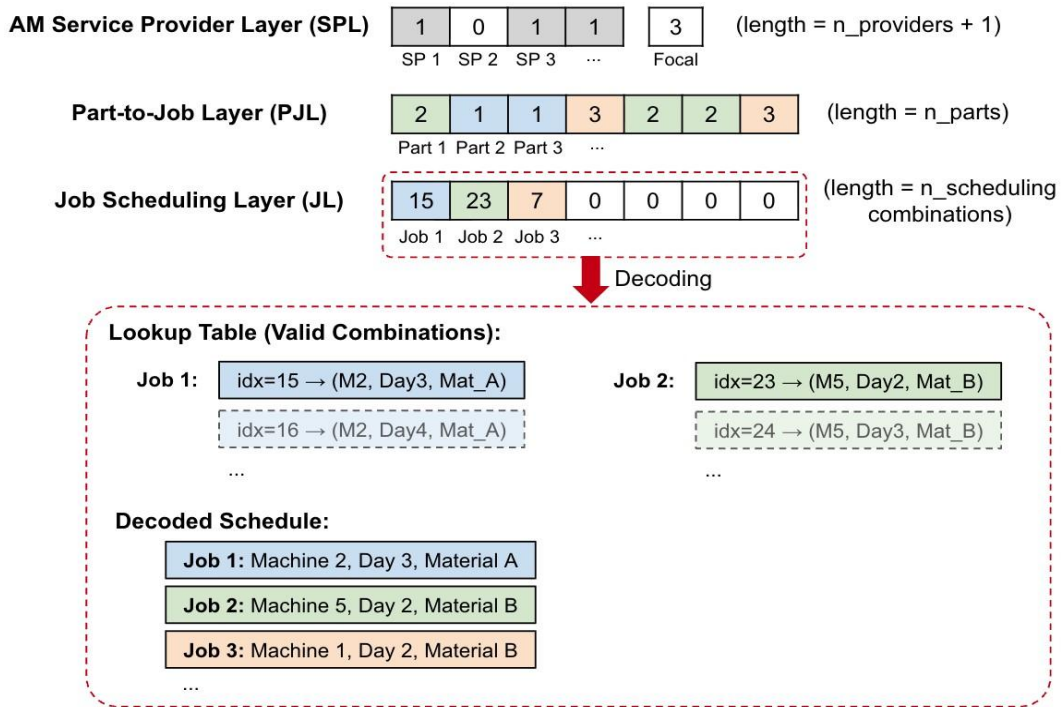


Figure 5. Example encoding.

5.2. Initialization

Initial solutions are generated using a cost-aware strategy to promote convergence efficiency while preserving population diversity. The initialization process is designed to avoid physically infeasible configurations.

In the SPL, service providers are randomly selected with the constraint that at least one provider must be included. The focal provider is preferentially chosen from among the selected providers that are eligible to serve as focal ($\kappa_k = 1$). If no selected provider satisfies this condition, one eligible provider is randomly added.

In the PJL, parts are grouped into jobs with an upper bound on the total number of jobs set to approximately half the number of parts, balancing utilization efficiency and capacity feasibility (e.g., 16 jobs for a 33-part instance).

In the JL, each job is assigned a valid combination index from the precomputed table, restricted to machines owned by the selected providers and satisfying material compatibility for all parts in the job.

Among feasible combinations, job assignments are determined using multi-criteria scoring: (1) time efficiency favors lower-cost machines and those already assigned to jobs, (2) material consolidation prioritizes materials already in use to reduce diversity, and (3) provider balance avoids underutilization while preferring cost-efficient machines. Scores are aggregated, and assignments are selected probabilistically from top-ranked candidates to preserve diversity. Jobs without parts are marked inactive, and providers without jobs (except focal) are removed.

After initialization, a repair procedure enforces structural feasibility: at least one provider selected, focal provider both selected and eligible, job-machine assignments consistent with ownership, and redundant providers removed (detailed in Section 5.6).

5.3. Fitness evaluation

The fitness function minimizes total production cost, consistent with the MILP objective (Equation (1)). To handle potential constraint violations after genetic operations, penalty terms proportional to the degree of violation are added to the cost. Penalties are applied for violations of machine capacity (build area and working time), material compatibility, and service provider operating hours.

5.4. Selection

A combination of elitism and tournament selection (Goldberg and Deb 1991) is adopted to evolve the population. Elitism ensures that a fixed proportion of the best-performing individuals is directly carried over to the next generation, preserving high-quality solutions. For the remaining individuals, tournament selection is applied, whereby a subset of candidates is randomly sampled and the individual with the best fitness is selected as a parent. The algorithm follows a generational replacement strategy, in which newly generated offspring replace the non-elite portion of the population. Specific parameter settings, including the elite ratio and tournament size, are reported in Section 6.2.

5.5. Genetic operators

5.5.1. Crossover operation

Since each layer exhibits distinct information structures and constraints, the corresponding genetic operators are tailored to best capture the characteristics of each layer (Zhang et al. 2022).

In the SPL, uniform crossover is applied to the binary provider selection genes. After crossover, feasibility is ensured by checking that at least one provider is selected. The focal provider is then updated by selecting from the chosen providers eligible to serve as focal; if no such provider exists, one eligible provider is randomly added to the selection.

In the PJI, uniform crossover is also used, where each part's job assignment is independently exchanged between parents. This allows flexible mixing of grouping patterns while preserving the independence of individual part assignments.

In the JL, single-point crossover is applied by randomly selecting a crossover point and swapping the segments beyond it. This maintains contiguous scheduling blocks while introducing diversity. Jobs with no assigned parts are marked as inactive (index -1) after crossover.

5.5.2. Mutation operation

As with crossover, mutation is applied independently to each chromosome layer, with minor adjustments to reduce reliance on the subsequent repair mechanism.

In the SPL, bit-flip mutation is used: the selection status (0 or 1) of a randomly chosen provider is flipped. Feasibility is maintained by ensuring that at least one provider remains selected, and the focal provider is updated accordingly to reflect the new selection.

In the PJP, a randomly selected part is reassigned to a different job. If the target job was previously inactive, the algorithm attempts to activate it by assigning a valid machine-day-material combination that is compatible with all parts in the job. If no valid combination exists, the job remains inactive.

In the JL, a randomly selected active job is reassigned to a new valid combination, considering only options involving machines from the selected providers and compatible materials. The selection process employs the same semi-greedy heuristic as in initialization to balance time efficiency and material consolidation.

5.6. Repair mechanism

Due to the multi-layer chromosome structure, infeasible or inconsistent solutions may arise both after initialization and after genetic operations. To maintain feasibility, a repair mechanism is applied immediately after initialization and after each genetic operation to restore structural consistency across layers while preserving the original solution characteristics as much as possible.

For the SPL, at least one provider must be selected. If crossover violates the focal provider eligibility requirement, a focal-eligible provider is randomly selected from the unselected pool and added to the set of selected providers.

For the JL, two violations are addressed. First, jobs with no assigned parts are marked inactive (index -1). Second, jobs assigned to machines from unselected providers or with invalid indices are repaired through reassignment, restricted to machines from selected providers and satisfying material compatibility. If no valid combination exists, the job is marked inactive. Reassignment follows the multi-criteria selection strategy from initialization.

After repairing SPL and JL, providers without machine utilization (except focal) are removed to eliminate redundancy.

The PJP does not require repair, as part-to-job assignments are inherently flexible.

6. Case-based experimental analysis

6.1. GA model performance comparison

First, the GA is validated by comparing its results with MILP solutions from Section 4.2 using the same dataset to examine whether the GA can reproduce the MILP solution while reducing computation time.

The GA identifies the optimal solution with a total cost of 46,500 KRW, which matches the MILP result obtained under a 3,600-second time limit, while reducing computation time to 72.7 seconds (approximately 50-fold improvement). These results confirm that the GA achieves equivalent solution quality with substantially improved computational efficiency, supporting its use for larger-scale instances. The cost breakdown further confirms this equivalence: both approaches yield identical allocations across all cost components, including basic order cost, part production cost, focal provider cost, and delivery cost.

Figure 6 compares the cost breakdown of the solution with that of the single-provider model of Kim and Jung (2026), expressed as the share of each component in total cost. In the single-provider model, part production cost accounts for 87.3% of total cost, with the remaining 12.7% attributed to basic order cost. In the multi-provider model, focal provider cost and delivery cost, which are absent in the single-provider setting, account for 10.8% (5,000 KRW) and 4.3% (2,000 KRW) of total cost, respectively. The relative share of part production cost decreases to 74.2%, while its absolute value remains unchanged at 34,500 KRW, indicating that the higher total cost of the multi-provider model reflects the additional coordination and delivery functions rather than reduced production efficiency.

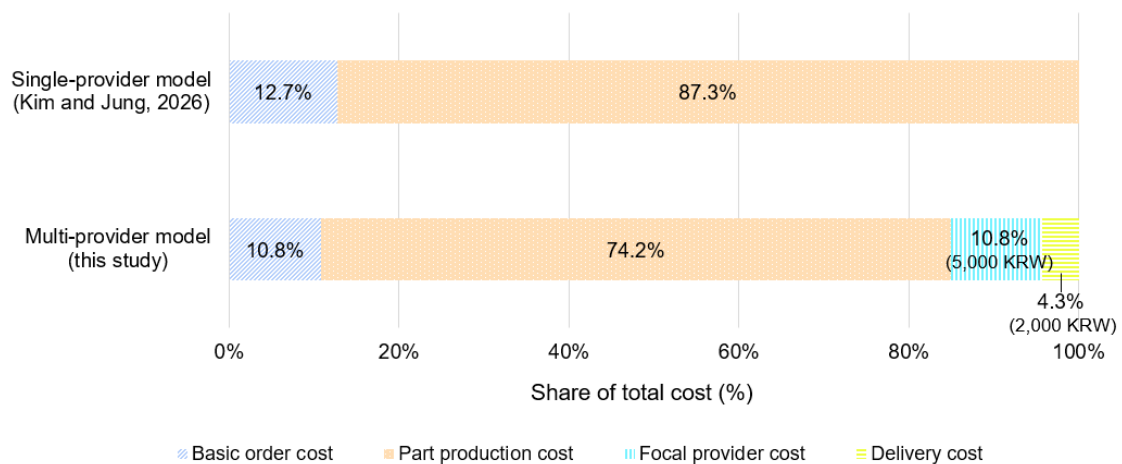


Figure 6. Cost breakdown comparison between the single-provider model (Kim and Jung 2026) and the proposed multi-provider model.

To further assess the computational performance of the proposed GA, the best-fit (BF) heuristic and benchmark instances from Li et al. (2017) were used for comparison. The BF heuristic addresses the problem of allocating parts to jobs and assigning jobs to multiple non-identical AM machines to minimize average production cost per unit volume, which shares the part-to-job grouping and multi-machine assignment decisions with PJJ and JL of the proposed framework. However, it differs from the current study in several key aspects: it considers a single provider environment without provider selection (SPL), omits daily scheduling and lead time constraints, assumes a single material type, and does not account for logistics costs. Despite these differences, Li et al. (2017) represents the most structurally similar study in the AM production planning literature, making it a suitable reference point for algorithmic comparison.

The benchmark instances include four problem sizes ranging from 10 to 200 parts across 2 to 6 machines, and an equivalent cost formulation was applied for fair comparison. The results show that the GA consistently achieved solutions within 0.76% of the BF heuristic across all instances, while requiring substantially less computation time. For example, computation time was reduced from 5,981 seconds to 832 seconds for the largest 200-part instance. These results are particularly noteworthy given that the proposed GA is optimized for a substantially more complex integrated problem than that of Li et al. (2017), yet still achieves competitive solution quality and computational efficiency on their benchmark instances.

6.2. Experimental design

The experimental environment is based on a Korean open-source platform for assistive devices operated by the Korea National Rehabilitation Center. Originally focused on sharing designs, the platform is evolving toward networked production where affiliated providers fabricate and deliver products (Jeong 2024; Kim and Jung 2026), providing a realistic shared AM environment.

The sample product is a writing assistive device for children with limited upper-limb mobility (Figure B1 in Appendix B), supporting stable writing and drawing for users who can move only their upper arms (The Korea National Rehabilitation Center, n.d.). It consists of ten components, of which three core AM-produced parts are considered in this analysis (Table B1 in Appendix B).

Figure 7 illustrates ten AM service providers and the customer in Incheon, South Korea. Providers are classified into four categories: assistive technology (AT) centers, makerspaces, 3D printing companies, and individual providers. Only AT centers are assumed to be able to serve as focal providers due to their expertise and after-service capability. The distances between providers and the customer are in Table B2 in Appendix B.

The basic operational information of the ten AM service providers is summarized in Table B3 (Appendix B), including provider type, basic ordering cost, focal service cost (where applicable), and operating hours. Detailed specifications and costs of the AM machines shared by these providers are reported in Table B4 (Appendix B). Machine ownership and cost information were incorporated from real-world data where available, and otherwise assumed based on typical provider characteristics.

The experiments were conducted using the same computing environment described in Section 4.2. For the MILP model, the computation time was capped at 3,600 seconds. The GA was implemented with a population size of 200 over 500 generations. The elite rate, crossover rate, and mutation rate were set to 0.05, 0.8, and 0.4, respectively, and tournament selection was applied with a tournament size of 2. Each setting was repeated five times with the best solution reported.

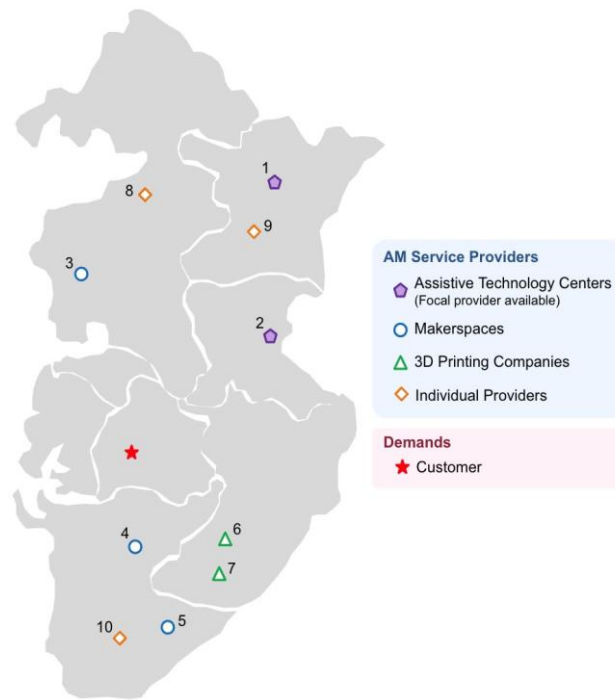


Figure 7. Geographic distribution of AM service providers and the customer in the experimental case study.

6.3. Computational performance comparison

The computational performance of the proposed GA-based approach is compared with that of the MILP model in terms of solution quality, measured by total cost, and computation time. Problem complexity is varied along two dimensions: (1) order size, represented by the number of parts, and (2) the number of available AM service providers. The comparison results are summarized in

In the first experiment, order size was increased from 3 to 45 parts while fixing the service provider to SP1, which owns a single AM machine. For small instances (P3, P6, P9, P15), both GA and MILP obtained identical optimal solutions, and MILP proved optimality within the 3,600-second time limit. For larger instances (P30 and P45), MILP reached the time limit without proving optimality, whereas GA obtained the same solutions within significantly shorter computation times (23–45 seconds), indicating near-optimal performance.

In the second experiment, the number of providers was varied while fixing the order size at 15 parts. As the number of providers increased, the solution space expanded rapidly, leading to greater computational difficulty. In all multi-provider instances, MILP reached the time limit without proving optimality. GA consistently produced high-quality solutions within short computation times. For example, in the SP3 (6 machines) case, GA achieved a 5.39% lower cost than MILP's best incumbent solution (281,000 vs. 297,000 KRW). For SP5 (15 machines), both methods converged to the same solution. In one case with SP10 (30 machines), MILP obtained a slightly lower cost (0.98% gap), although optimality was not proven by either method.

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Table 4. Comparison of MILP and GA results under varying number of parts and AM service providers.

Instance	MILP	MILP time(s)	GA	GA time(s)	Gap
<i>Varying number of parts</i>					
P3 SP1(M1)	75,000*	0.08	75,000	6.62	0.00%
P6 SP1(M1)	123,000*	1.06	123,000	7.55	0.00%
P9 SP1(M1)	171,000*	548	171,000	8.89	0.00%
P15 SP1(M1)	281,000*	585	281,000	12.86	0.00%
P30 SP1(M1)	535,000	3,600	535,000	23.39	0.00%
P45 SP1(M1)	789,000	3,600	789,000	44.68	0.00%
<i>Varying number of AM service providers</i>					
P15 SP1(M1)	281,000*	585	281,000	13.06	0.00%
P15 SP2(M3)	281,000	3,600	281,000	16.05	0.00%
P15 SP3(M6)	297,000	3,600	281,000	19.89	-5.39%
P15 SP5(M15)	207,500	3,600	207,500	92.57	0.00%
P15 SP10(M30)	152,600	3,600	154,100	129.73	0.98%

*Optimal MILP result obtained before time limit.

These results highlight the practical advantages of the GA over the MILP model for larger-scale instances. MILP guarantees optimality for small instances (up to 15 parts, single provider) while degrading rapidly as complexity increases. GA consistently produces high-quality

solutions within practical computation times across all tested scales, matching or outperforming MILP in most cases, making it a robust approach for SCCP problems at scales relevant to shared AM environments.

6.4. Supply Chain Configuration and Planning (SCCP) Analysis

This section examines how SCCP decisions respond to variations in order characteristics in shared AM environments, focusing on changes in provider selection, machine allocation, and production planning.

Although SCCP outcomes may be influenced by various order attributes, such as material requirements, geometric complexity, and lead time constraints, analyzing all factors simultaneously would obscure their individual effects. To enable a systematic analysis, this study focuses on two representative characteristics: order size, which determines production capacity requirements, and lead time, which reflects fulfillment urgency.

Accordingly, SCCP decisions are evaluated across combinations of order sizes (3, 6, 9, 15, 30, 45, 60, 80, 100, and 120 parts) and lead time limits ranging from 5 days to 1 day, while other characteristics are held constant. This two-dimensional experimental design allows the effects of order scale and time pressure on supply chain configuration to be examined in a structured manner.

Table 5 summarizes the computational performance comparison. For instances up to 30 parts, both methods yield comparable solution quality with optimality gaps below 1%. Beyond this range, MILP performance deteriorates rapidly; for the 45-part instance, GA obtained a 31.05% better solution, and MILP failed to find any feasible solution for instances exceeding 45 parts. GA successfully generated solutions across all tested sizes, and is used for subsequent SCCP analysis.

Table 5. Comparison of MILP and GA results under varying number of parts with 10 AM service providers.

Instance	MILP	MILP time(s)	GA	GA time(s)	Gap
P3 SP10(M30)	46,500*	8.38	46,500	37.48	0.00%
P6 SP10(M30)	63,000	3,600	63,000	52.62	0.00%
P9 SP10(M30)	93,100	3,600	93,100	73.48	0.00%
P15 SP10(M30)	152,600	3,600	154,100	97.57	0.98%
P30 SP10(M30)	326,200	3,600	326,900	248.46	0.21%
P45 SP10(M30)	734,400	3,600	506,400	503.73	-31.05%
P60 SP10(M30)	No feasible solution	3,600	691,400	559.51	-
P80 SP10(M30)	No feasible solution	3,600	954,600	911.9	-

P100 SP10(M30)	No feasible solution	3,600	1,305,600	1,086.18	-
P120 SP10(M30)	No feasible solution	3,600	1,693,700	1,854.97	-

*Optimal MILP result obtained before time limit.

6.4.1. Impact of order size

Table 6 summarizes how SCCP decisions evolve as order size increases from 3 to 120 parts, including the number of participating providers, selected machines, and resulting lead times.

Three consistent patterns emerge as order size increases. First, SP2 is selected as the focal provider across most order sizes (3-80 parts), while SP1 takes the role for very large orders (100-120 parts). Although SP1 offers a lower focal service fee (3,000 vs. 5,000 KRW), SP2 is generally preferred due to its more flexible and cost-efficient manufacturing capability. SP2's moderately sized machines (M2, M3) offer lower operating costs and its central location (Figure) reduces transportation distances.

For very large orders (100–120 parts), however, the cost-driven selection logic shifts toward throughput capacity. Meeting the 5-day lead time constraint requires machines capable of handling high production volumes efficiently, rather than simply minimizing operating costs. Accordingly, SP1 emerges as the focal provider: despite its higher operating costs, its machine (M1) offers a substantially larger build area (3,900 cm²) and faster forming time (0.027 hr/cm³), making it better suited to fulfill large-volume orders within the given time constraint.

Table 6. Summary of supply chain configurations and lead times for different order sizes.

Order size (parts)	Number of AM service providers	Selected AM service providers	Number of AM machines used	Lead time (days)
3	2	SP2*, SP10	2	2.30
6	2	SP2*, SP10	2	4.30
9	3	SP2*, SP9, SP10	4	4.30
15	4	SP2*, SP4, SP9, SP10	5	4.30
30	5	SP2*, SP5, SP8, SP9, SP10	6	4.53
45	6	SP2*, SP4, SP5, SP8, SP9, SP10	9	4.53
60	6	SP2*, SP4, SP5, SP8, SP9, SP10	11	4.53
80	7	SP2*, SP4, SP5, SP7, SP8, SP9, SP10	12	4.53

100	7	SP1*, SP4, SP5, SP7, 13 SP8, SP9, SP10	4.55
120	8	SP1*, SP4, SP5, SP6, 15 SP7, SP8, SP9, SP10	4.55

* Focal AM service provider

Second, the supply chain expands progressively with order size following a cost-driven provider selection pattern (Figure 8). Small orders (3–6 parts) involve only two providers, while additional providers are incorporated as capacity requirements increase. Individual providers (SP8–SP10), which offer the most cost-efficient capacity, are selected first, followed by makerspaces (SP3–SP5) for medium-sized orders. Specialized 3D printing companies (SP6, SP7) are not selected for orders up to 60 parts, as their higher prices are not justified when slower, lower-cost providers can still meet delivery requirements. However, as order volume grows substantially, they are incorporated to support production capacity; SP7 is selected from 80-part instance onward, and SP6 is additionally included for the 120-part instance.

Third, this cost-oriented logic is reflected at the machine level, where utilization is measured as the ratio of actual to available time, averaged by machine types: economical machines (Crealty Ender-5 and Cubicon Single Plus) and industrial machines (all remaining models). For small orders, economical machine utilization varies depending on the number of selected providers and machines. From the 30-part instance onward, however, it stabilizes at a consistently high level (58.2–73.5%), reflecting their role as the primary source of production capacity throughout the supply chain.

Industrial machines, in contrast, show a general increase in utilization as order size grows (23.5% for the 30-part instance, up to 63.0% for the 100-part instance), indicating that higher-capacity equipment becomes increasingly necessary to meet lead time constraints under substantial production volumes. These results indicate that machine-level scheduling decisions reinforce cost-driven provider selection under moderate order volume, while shifting toward a capacity-driven allocation as order size grows substantially.

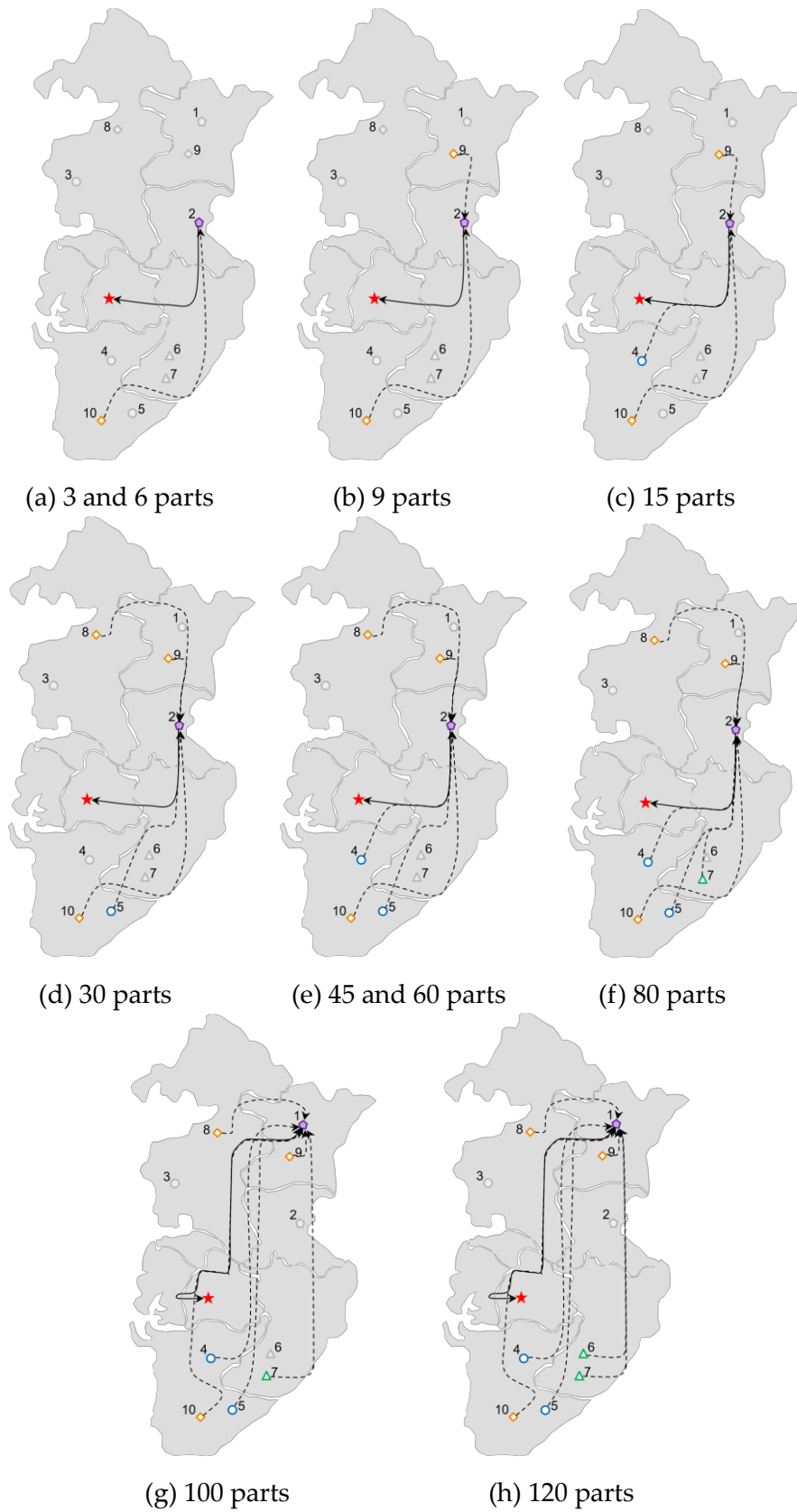


Figure 8. Supply chain configurations under order size changes: (a) 3 and 6 parts, (b) 9 parts, (c) 15 parts, (d) 30 parts, (e) 45 and 60 parts, (f) 80 parts, (g) 100 parts, (h) 120 parts.

6.4.2. Impact of lead time requirement

This section examines how tightening lead time constraints alters SCCP decisions. In customization contexts, customer preferences are heterogeneous, with some prioritizing cost minimization while others emphasizing shorter lead times (Tookanlou and Wong 2020; Dolan and Henderson 2014; Larsson Ranada and Lidström 2019). To isolate the effect of time pressure, SCCP configurations and production plans are analyzed under five lead time settings: 5 days (baseline), 4 days, 3 days, 2 days, and 1 day, while fixing the order size at 15 parts.

Figure 9 shows the resulting configurations. When the lead time is reduced from 5 days to 3 days, the provider set remains unchanged, consisting of SP2 as the focal provider together with SP4, SP9, and SP10. Under the 2-day constraint, SP5 is additionally incorporated to provide extra capacity. A substantial configuration shift occurs under the 1-day requirement, where SP1 becomes the focal provider and SP6, a specialized 3D printing company, is newly selected.

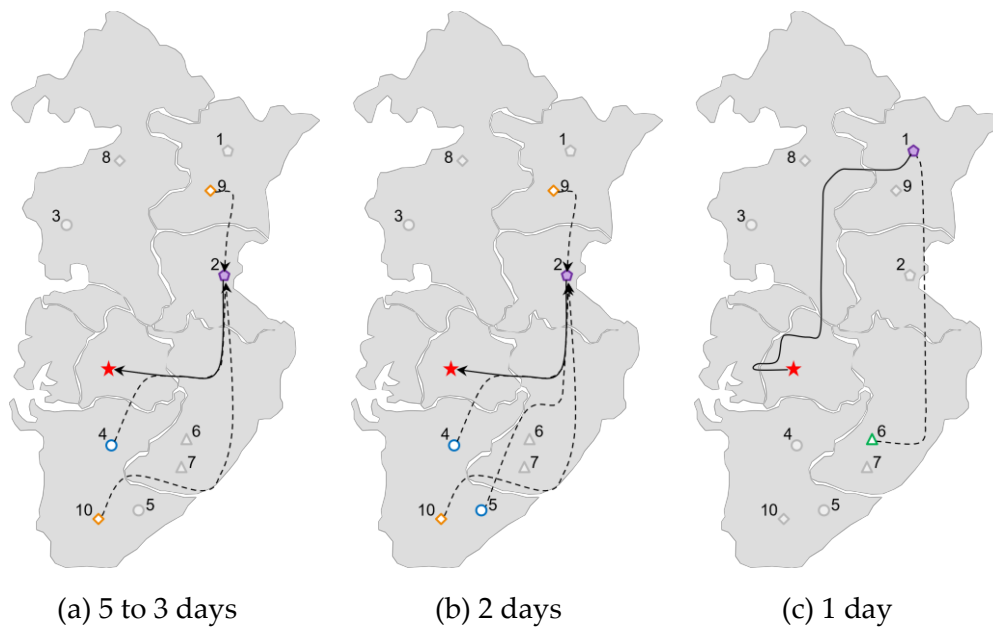


Figure 9. Supply chain configurations under lead time requirement changes: (a) 5 to 3 days, (b) 2 days, and (c) 1 day.

Under the 1-day constraint, production shifts toward high-performance machines with lower forming time and larger capacity (

Table 7), with schedule nearly exhausting providers' daily operating limits.

The economic impact of these configuration changes is substantial (Figure 10). Under relaxed constraints (5–3 days), total costs increase moderately from 154,100 to 176,600 KRW. In contrast, the 1-day lead time results in a sharp cost increase to 379,100 KRW, representing a 146% premium over the baseline. This escalation reflects the need to employ premium-priced specialized providers and high-performance machines to meet extreme time pressure, indicating that stringent lead time requirements fundamentally override cost-oriented SCCP decisions in shared AM environments.

Table 7. Comparison of production plans under 5-day and 1-day lead time constraints (15 parts).

Lead time limit	AM service provider	AM machine	Day	Job	Part(s)	Processing time
5 days	SP2*	M2	D1	J15	P15	3.14h
	SP4	M7	D1	J1	P4	14.35h
	SP9	M28	D1	J10	P5	22.12h
			D2	J11	P14	22.12h
			D3	J8	P8	22.12h
			D1	J13	P2	22.12h
	SP10	M29	D2	J12	P11	22.12h
			D1	J5	P9	5.59h
				J9	P7	14.35h
			D2	J4	P12, P13	19.76h
			D3	J2	P3, P10	19.76h
			D4	J7	P6	5.59h
				J3	P1	14.35h
1 day	SP1*	M1	D1	J1	P5, P14	5.33h
	SP6	M19	D1	J3	P2, P6, P10, P11, P15	6.50h
		M20	D1	J4	P4, P7, P8, P9	6.95h
		M21	D1	J2	P1, P2, P12, P13	6.95h

* Focal AM service provider

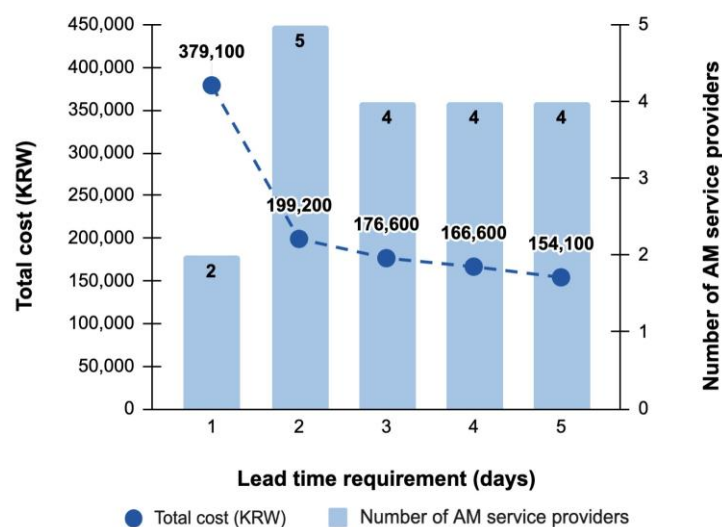


Figure 10. Total cost and number of AM service providers under varying lead time requirements.

7. Discussion

The proposed framework establishes the computational feasibility of on-demand SCCP in shared AM environments. However, translating this capability into operational practice requires enabling conditions that are not yet fully established in the industry. Drawing on the possibility theory of Pil et al. (2024), the on-demand SCCP approach presented in this study represents an *available possibility*: theoretically grounded and computationally supported, yet constrained by structural conditions that inhibit practical realization.

Among the enabling conditions that remain underdeveloped, two are particularly salient for the actualization of on-demand SCCP.

First, the framework relies on timely access to provider- and machine-level information. A baseline set of static information, including machine specifications, material compatibility, pricing policies, and operating schedules, must be consistently maintained and shared across platform participants. Beyond this, real-time information on machine-level availability is required at the moment of order arrival to enable supply chain configuration (Wang et al. 2021). Once production is underway, ongoing visibility into actual job completion times across providers is further needed to ensure coordinated execution. While these requirements may appear modest individually, their simultaneous fulfillment across a dynamically assembled provider network represents a significant infrastructure challenge for current shared AM platforms.

Second, the multi-provider structure inherent to on-demand SCCP raises unresolved questions of inter-organizational quality assurance. Although a focal provider assumes responsibility for assembly and final delivery, ensuring the quality of parts produced by different providers remains structurally ambiguous. Unlike traditional supply chains where quality assurance norms develop through repeated interaction (Gulati 1995), on-demand configurations offer no such opportunity for trust accumulation. Platform-level mechanisms, such as provider performance tracking systems, quality verification protocols, or contractual frameworks governing inter-provider responsibilities, are needed to substitute for relationship-based trust in transient provider networks.

8. Conclusion

This study addressed the on-demand SCCP problem in shared AM environments from a platform operation perspective. Unlike traditional supply chain design problems that focus on long-term strategic decisions, shared AM platforms require order-level supply chain configuration and planning based on real-time order characteristics. To support this need, an integrated optimization framework was proposed that jointly determines service provider selection and production planning decisions.

The problem was formulated as a MILP model capturing key operational features of shared AM environments, including machine heterogeneity, material compatibility, capacity constraints, and provider-specific pricing and operating schedules. To handle computational challenges in practical-scale instances, a GA-based solution approach with a multi-layer chromosome structure was developed. Computational results showed that the GA achieved high-quality solutions within practical computation times, including instances where the MILP became intractable.

Experimental analyses revealed systematic patterns in SCCP decisions. Under relaxed lead time requirements, supply chain configurations followed a cost-driven structure, with gradual expansion as order size increased. In contrast, stringent lead time constraints induced a shift toward faster and more expensive providers, leading to substantial cost increases. These findings highlight how order characteristics fundamentally shape supply chain configuration and planning decisions in shared AM environments.

The findings provide managerial insights for AM service providers participating in shared AM platforms. Providers operating mid- to low-performance AM equipment benefit from cost-efficient operations, as SCCP decisions under relaxed lead time constraints favor lower operating costs. The influence of geographic location depends on the platform's cost structure; when delivery costs are modest relative to production costs, machine operating costs dominate configuration decisions, allowing peripheral providers to remain competitive. Accordingly, such providers should align pricing and participation strategies with their cost efficiency and platform-specific cost structure.

Providers offering high-performance AM capabilities are rarely selected under cost-driven conditions for small to medium-sized orders, but become increasingly relevant as order volume grows substantially, and critical under stringent lead time requirements. Rather than competing on price for routine orders, these providers are better positioned to target time-sensitive demand through premium services. Given their limited selection frequency, participation in shared AM platforms should be viewed as a supplementary utilization opportunity.

This study has several limitations that suggest directions for future research. First, the proposed framework derives SCCP decisions based on nominal provider availability and machine schedules at order placement. In practice, execution uncertainties from prior orders, such as print failures, machine breakdowns, or rework, may cause deviations between assumed and actual operating conditions. Future research could incorporate execution variability to enhance robustness under uncertainty.

Second, although the model captures diverse operational attributes, quality differences across providers and machines are not explicitly considered. In particular, quality consistency across parts produced by multiple providers is not enforced, which may lead to assembly or performance issues. Future studies could incorporate quality attributes or consistency constraints.

Finally, the framework does not leverage historical execution data from prior orders. As shared AM platforms mature, learning from accumulated execution records, such as delays or machine reliability, could improve parameter estimation. Future research may explore learning-enhanced SCCP approaches using historical execution data.

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Disclosure statement

No potential conflict of interest was reported by the authors.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon request.

Declarations - Generative AI statement

Portions of the manuscript text were refined for English grammar and clarity using ChatGPT (GPT-5.2 Thinking, OpenAI). The tool was used solely for language editing and did not generate new scientific content, data, results, or interpretations. All revisions were reviewed and approved by the authors.

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References

1. Akmal, Jan Sher, Mika Salmi, Roy Björkstrand, Jouni Partanen, and Jan Holmström. 2022. "Switchover to Industrial Additive Manufacturing: Dynamic Decision-Making for Problematic Spare Parts." *International Journal of Operations & Production Management* 42 (13): 358–384. <https://doi.org/10.1108/IJOPM-01-2022-0054>.
2. Atzeni, Eleonora, Luca Iuliano, Paolo Minetola, and Alessandro Salmi. 2010. "Redesign and Cost Estimation of Rapid Manufactured Plastic Parts." *Rapid Prototyping Journal* 16 (5): 308–317. <https://doi.org/10.1108/13552541011065704>.
3. Barz, Andreas, Tobias Buer, and Hans-Dietrich Haasis. 2016. "Quantifying the Effects of Additive Manufacturing on Supply Networks by Means of a Facility Location-Allocation Model." *Logistics Research* 9 (1): 13. <https://doi.org/10.1007/s12159-016-0140-0>.
4. Chowdhury, Sudipta, Omid Shahvari, Mohammad Marufuzzaman, Jack Francis, and Linkan Bian. 2019. "Sustainable Design of On-Demand Supply Chain Network for Additive Manufacturing." *IIE Transactions* 51 (7): 744–765. <https://doi.org/10.1080/24725854.2018.1532134>.
5. Cuellar, Juan Sebastian, Gerwin Smit, Dick Plettenburg, and Amir Zadpoor. 2018. "Additive Manufacturing of Non-Assembly Mechanisms." *Additive Manufacturing* 21: 150–158. <https://doi.org/10.1016/j.addma.2018.02.004>.
6. Dolan, Michael John, and Graham Iain Henderson. 2014. "Patient and Equipment Profile for Wheelchair Seating Clinic Provision." *Disability and Rehabilitation: Assistive Technology* 9 (2): 136–143. <https://doi.org/10.3109/17483107.2013.807441>.
7. Gibson, Ian, David W. Rosen, and Brent Stucker. 2010. *Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing*. Boston, MA: Springer US. <https://doi.org/10.1007/978-1-4419-1120-9>.
8. Goldberg, David E., and Kalyanmoy Deb. 1991. "A Comparative Analysis of Selection Schemes Used in Genetic Algorithms." In *Foundations of Genetic Algorithms*, 1:69–93. Elsevier. <https://doi.org/10.1016/B978-0-08-050684-5.50008-2>.
9. Gulati, Ranjay. 1995. "Does Familiarity Breed Trust? The Implications of Repeated Ties for Contractual Choice in Alliances." *Academy of Management Journal* 38 (1): 85–112. <https://doi.org/10.5465/256729>.
10. Hedenstierna, Carl Philip T., Stephen M. Disney, Daniel R. Eyers, Jan Holmström, Aris A. Syntetos, and Xun Wang. 2019. "Economies of Collaboration in Build-to-model Operations." *Journal of Operations Management* 65 (8): 753–773. <https://doi.org/10.1002/joom.1014>.
11. Hopkinson, Neil, and Philip Dickens. 2003. "Analysis of Rapid Manufacturing—Using Layer Manufacturing Processes for Production." *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science* 217 (1): 31–39. <https://doi.org/10.1243/095440603762554596>.
12. Huang, Samuel H., Peng Liu, Abhiram Mokasdar, and Liang Hou. 2013. "Additive Manufacturing and Its Societal Impact: A Literature Review." *The International Journal of Advanced Manufacturing Technology* 67 (2013): 1191–1203. <https://doi.org/10.1007/s00170-012-4558-5>.
13. ISO/ASTM. 2021. *ISO/ASTM 52900:2021 – Additive Manufacturing – General Principles – Fundamentals and Vocabulary*. <https://doi.org/10.1520/F2792-12A.2>.

14. Jeong, Gwang-seong. 2024. "Strengthening Community-Based Rehabilitation for the Disabled." *Medical News*, April 4, 2024. <http://www.bosa.co.kr/news/articleView.html?idxno=2220084>.
15. Jimo, Ajeseun, Christos Braziotis, Helen Rogers, and Kulwant Pawar. 2022. "Additive Manufacturing: A Framework for Supply Chain Configuration." *International Journal of Production Economics* 253: 108592. <https://doi.org/10.1016/j.ijpe.2022.108592>.
16. Khajavi, Siavash H., Jan Holmström, and Jouni Partanen. 2018. "Additive Manufacturing in the Spare Parts Supply Chain: Hub Configuration and Technology Maturity." *Rapid Prototyping Journal* 24 (7): 1178–1192. <https://doi.org/10.1108/RPJ-03-2017-0052>.
17. Khajavi, Siavash H., Jouni Partanen, and Jan Holmström. 2014. "Additive Manufacturing in the Spare Parts Supply Chain." *Computers in Industry* 65 (1): 50–63. <https://doi.org/10.1016/j.compind.2013.07.008>.
18. Kim, Songi, and Hosang Jung. 2026. "Service Provider Recommendation System in Shared Additive Manufacturing: A Case Study on Assistive Technology Platform in South Korea." *International Journal of Production Research* 64 (10): 4158–4180. <https://doi.org/10.1080/00207543.2025.2539309>.
19. Kim, Songi, and Keeheon Lee. 2023. "The Paradigm Shift of Mass Customisation Research." *International Journal of Production Research* 61 (10): 3350–3376. <https://doi.org/10.1080/00207543.2022.2081629>.
20. Kucukkoc, Ibrahim. 2019. "MILP Models to Minimise Makespan in Additive Manufacturing Machine Scheduling Problems." *Computers & Operations Research* 105: 58–67. <https://doi.org/10.1016/j.cor.2019.01.006>.
21. Kucukkoc, Ibrahim. 2024. "Scheduling of Distributed Additive Manufacturing Machines Considering Carbon Emissions." *An International Journal of Optimization and Control: Theories & Applications* 14 (1): 20–31. <https://doi.org/10.11121/ijocta.1444>.
22. Kunovjanek, Maximilian, Nils Knofius, and Gerald Reiner. 2022. "Additive Manufacturing and Supply Chains – a Systematic Review." *Production Planning & Control* 33 (13): 1231–1251. <https://doi.org/10.1080/09537287.2020.1857874>.
23. Larsson Ranada, Åsa, and Helene Lidström. 2019. "Satisfaction with Assistive Technology Device in Relation to the Service Delivery Process-A Systematic Review." *Assistive Technology: The Official Journal of RESNA* 31 (2): 82–97. <https://doi.org/10.1080/10400435.2017.1367737>.
24. Li, Qiang, Ibrahim Kucukkoc, and David Z. Zhang. 2017. "Production Planning in Additive Manufacturing and 3D Printing." *Computers & Operations Research* 83: 157–172. <https://doi.org/10.1016/j.cor.2017.01.013>.
25. Li, Xueping, and Kaike Zhang. 2018. "Single Batch Processing Machine Scheduling with Two-Dimensional Bin Packing Constraints." *International Journal of Production Economics* 196 (February 2018): 113–121. <https://doi.org/10.1016/j.ijpe.2017.11.015>.
26. Li, Yao, Yang Cheng, Qing Hu, Shenghan Zhou, Lei Ma, and Ming K. Lim. 2019. "The Influence of Additive Manufacturing on the Configuration of Make-to-Order Spare Parts Supply Chain under Heterogeneous Demand." *International Journal of Production Research* 57 (11): 3622–3641. <https://doi.org/10.1080/00207543.2018.1543975>.
27. Liu, Peng, Samuel H. Huang, Abhiram Mokasdar, Heng Zhou, and Liang Hou. 2014. "The Impact of Additive Manufacturing in the Aircraft Spare Parts Supply Chain: Supply Chain Operation Reference (Scor) Model Based Analysis." *Production Planning & Control* 25 (13–14): 1169–1181. <https://doi.org/10.1080/09537287.2013.808835>.

28. Lupi, Francesco, Mario G. C. A. Cimino, Tomaž Berlec, Federico A. Galatolo, Marko Corn, Nejc Rožman, Andrea Rossi, and Michele Lanzetta. 2023. "Blockchain-Based Shared Additive Manufacturing." *Computers & Industrial Engineering* 183: 109497. <https://doi.org/10.1016/j.cie.2023.109497>.
29. Mellor, Stephen, Liang Hao, and David Zhang. 2014. "Additive Manufacturing: A Framework for Implementation." *International Journal of Production Economics* 149 (2014): 194–201. <https://doi.org/10.1016/j.ijpe.2013.07.008>.
30. Melo, M. T., S. Nickel, and F. Saldanha-da-Gama. 2009. "Facility Location and Supply Chain Management – A Review." *European Journal of Operational Research* 196 (2): 401–412. <https://doi.org/10.1016/j.ejor.2008.05.007>.
31. Nazir, Aamer, Ozkan Gokcekaya, Kazi Md Masum Billah, Onur Ertugrul, Jingchao Jiang, Jiayu Sun, and Sajjad Hussain. 2023. "Multi-Material Additive Manufacturing: A Systematic Review of Design, Properties, Applications, Challenges, and 3D Printing of Materials and Cellular Metamaterials." *Materials & Design* 226: 111661. <https://doi.org/10.1016/j.matdes.2023.111661>.
32. Petch, Michael. 2025. "2025 Wohlers Report: Exclusive Insights and Reader Discount." *3D Printing Industry*, April 10, 2025. <https://3dprintingindustry.com/news/2025-wohlers-report-exclusive-insights-and-reader-discount-238255/>.
33. Pil, Frits K., Stephen M. Disney, Jan Holmström, Benn Lawson, and Christopher Tang. 2024. "Possibility Theory: A Foundation for Theoretical and Empirical Explorations of Uncertainty." *Journal of Operations Management* 70 (8): 1182–1193. <https://doi.org/10.1002/joom.1341>.
34. Savolainen, Jyrki, and Mikael Collan. 2020. "How Additive Manufacturing Technology Changes Business Models? – Review of Literature." *Additive Manufacturing* 32 (March 2020): 101070. <https://doi.org/10.1016/j.addma.2020.101070>.
35. Simchi-Levi, David, Philip Kaminsky, and Edith Simchi-Levi. 2004. *Managing the Supply Chain: The Definitive Guide for the Business Professional*. New York. McGraw-Hill. <https://www.amazon.com/Managing-Supply-Chain-Definitive-Professional/dp/0071410317>.
36. The Korea National Rehabilitation Center. n.d. *Assistive Technology Open Platform*. Accessed October 18, 2024. http://www.nrc.go.kr/at_rd/.
37. Tookanolou, Parisa Bagheri, and Hartanto Wong. 2020. "Determining the Optimal Customization Levels, Lead Times, and Inventory Positioning in Vertical Product Differentiation." *International Journal of Production Economics* 221: 107479. <https://doi.org/10.1016/j.ijpe.2019.08.014>.
38. Tziantopoulos, Konstantinos, Naoum Tsolakis, Dimitrios Vlachos, and Loukas Tsironis. 2019. "Supply Chain Reconfiguration Opportunities Arising from Additive Manufacturing Technologies in the Digital Era." *Production Planning & Control* 30 (7): 510–521. <https://doi.org/10.1080/09537287.2018.1540052>.
39. Vidal, Carlos J., and Marc Goetschalckx. 1997. "Strategic Production-Distribution Models: A Critical Review with Emphasis on Global Supply Chain Models." *European Journal of Operational Research* 98 (1): 1–18. [https://doi.org/10.1016/S0377-2217\(97\)80080-X](https://doi.org/10.1016/S0377-2217(97)80080-X).
40. Walter, Manfred, Jan Holmström, and Hannu Yrjölä. 2004. "Rapid Manufacturing and Its Impact on Supply Chain Management." Paper presented at Logistics Research

- Network Annual Conference. *Proceedings of Logistics Research Network Annual Conference*, September 9, 2004.
41. Wang, Gang, Geng Zhang, Xin Guo, and Yingfeng Zhang. 2021. "Digital Twin-Driven Service Model and Optimal Allocation of Manufacturing Resources in Shared Manufacturing." *Journal of Manufacturing Systems* 59: 165–179. <https://doi.org/10.1016/j.jmsy.2021.02.008>.
 42. Wen, Xiaoyu, Yunjie Qian, Xiaonan Lian, Yuyan Zhang, Haoqi Wang, and Hao Li. 2023. "Improved Genetic Algorithm Based on Multi-Layer Encoding Approach for Integrated Process Planning and Scheduling Problem." *Robotics and Computer-Integrated Manufacturing* 84 (2023): 102593. <https://doi.org/10.1016/j.rcim.2023.102593>.
 43. Xu, Xinglu, Mark D. Rodgers, and Weihong (Grace) Guo. 2021. "Hybrid Simulation Models for Spare Parts Supply Chain Considering 3D Printing Capabilities." *Journal of Manufacturing Systems* 59: 272–282. <https://doi.org/10.1016/j.jmsy.2021.02.018>.
 44. Yang, Hui, Ruimin Chen, and Soundar Kumara. 2021. "Stable Matching of Customers and Manufacturers for Sharing Economy of Additive Manufacturing." *Journal of Manufacturing Systems* 61: 288–299. <https://doi.org/10.1016/j.jmsy.2021.09.013>.
 45. Yu, Chunyang, Xun Xu, Shiqiang Yu, Zhiqian Sang, Cheng Yang, and Xuanlin Jiang. 2020. "Shared Manufacturing in the Sharing Economy: Concept, Definition and Service Operations." *Computers & Industrial Engineering* 146: 106602. <https://doi.org/10.1016/j.cie.2020.106602>.
 46. Zehetner, Dominik, and Margaretha Gansterer. 2022. "The Collaborative Batching Problem in Multi-Site Additive Manufacturing." *International Journal of Production Economics* 248: 108432. <https://doi.org/10.1016/j.ijpe.2022.108432>.
 47. Zhang, Xu, Zhixue Liao, Lichao Ma, and Jin Yao. 2022. "Hierarchical Multistrategy Genetic Algorithm for Integrated Process Planning and Scheduling." *Journal of Intelligent Manufacturing* 33 (1): 223–246. <https://doi.org/10.1007/s10845-020-01659-x>.

Appendix A. Indices, parameters, and decision variables

Table A1. Indices.

i	Part, $i \in \{1, \dots, I\}$
j	Job, $j \in \{1, \dots, J\}$
k	AM service provider
m	AM machine in the shared AM platform
t	Time period (day), $t \in \{1, \dots, T\}$
r	Raw material, $r \in \{1, \dots, R\}$

Table A2. Parameters.

κ_k	Whether provider k can act as the focal service provider (binary)
σ_k	Accumulated rating score of provider k (e.g., in a sharing economy context), ($0 \leq \sigma_k \leq 1$)
ψ	Minimum average rating threshold for the providers in the supply chain ($0 \leq \psi \leq 1$)
c_k^{order}	Basic ordering cost of provider k
c_m^{basic}	Machine usage basic fee of machine m
c_m^{extra}	Machine usage extra fee of machine m per hour
$c_{\square}^{\text{del}}$	Unit delivery cost of logistics
c_k^{rm}	Extra material handling cost of provider k
c_k^{foc}	Additional cost incurred if provider k acts as the focal provider (charging for assembly and final product preparation)
s_m	Setup time of machine m
v_i	Volume of part i
b_i	Base area of part i
β_m	Base area of the build chamber of machine m
δ_m	Forming time of machine m for a unit volume
t^{post}	Post-processing time for the parts
t^{asby}	Assembly time
h_{mt}	Machine available time of machine m on day t (integer)
o_{kt}	Opening hours of provider k on day t (integer)
τ_m^{AM}	Minimum usage time threshold of machine m (integer)
ϕ_{kt}	Whether the day t is an opening day for provider k (binary)
d_{kt}^{next}	Next working day of the day t for provider k (integer, ≥ 1)
$d_{k,k'}$	Distance between providers k and k'
d_k^{cust}	Distance between provider k and the customer
v	Average speed of the delivery (km/h)
α_{ir}	Whether raw material r can be used to manufacture part i or not (binary) ($\sum_r \alpha_{ir} \geq 1$)
γ_{mr}	Whether machine m can handle raw material r or not (binary) ($\sum_r \gamma_{mr} \geq 1$)
u_{mk}	Whether machine m belongs to provider k (binary)

Table A3. Decision variables.

X_{ij}	Whether part i is assigned to job j or not (binary)
Y_{jmt}	Whether job j is processed by machine m on day t or not (binary)
R_{jmtr}	Whether job j is processed by machine m on day t with raw material r or not (binary)
Z_k^{foc}	Whether provider k is selected as the focal service provider of the supply chain
Z_k^{SP}	Whether service provider k takes a part of production in the supply chain
P_{jmt}	Processing time of job j at machine m on day t
S_{kt}	Whether day t is the starting day of the final production job for provider k (binary)
K_{mt}^{AM}	Whether machine m is used on day t or not (binary)
K_{kt}^{SP}	Whether service provider k takes a part manufacturing process on day t or not (binary)
T_{mt}	Working hour of machine m on day t
T_{mt}^{hour}	Working hour of machine m on day t (integer)
T_{mt}^{extra}	Excess usage time of machine m on day t beyond the threshold τ_m^{AM} (integer)
T_{mt}^{long}	The longest part fabrication time of machine m on day t
$T_{kt}^{\text{long, day}}$	The longest working hour among the machines of service provider k on day t
T_k^{part}	Completion time of part production at provider k
T_k^{Arrival}	Arrival time of part supplement from provider k
T^{Final}	Completion time of finishing all products for the order
Ω_{kr}	Whether material r is used in the production process of service provider k (binary)
N_k^{rm}	Number of material types used in provider k
C_{mt}^{AM}	Machine usage cost of machine m on day t
C_k^{SP}	Total cost of provider k
D_{kt}^{extra}	Whether extra working days are required for provider k after the makespan to finish the production (binary)

Appendix B. Dataset of main experiment



Figure B1. Sample product (The Korea National Rehabilitation Center, n.d.).

Table B1. Part information.

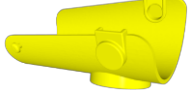
Part number	1	2	3
Name	Pen holder	Lower cover	Upper cover
Image			
Size (w×d×h, mm)	122×146×85	87×157×80	86×62×84
Volume (mm ³)	94,517	84,751	36,102
Base area (mm ²)	17,812	13,659	5,332

Table B2. Distance matrix between AM service providers and customer (unit: km).

		Providers										Customer
		1	2	3	4	5	6	7	8	9	10	
Providers	1	-	12	13	23	27	21	17	7	5	27	20
	2	12	-	20	15	16	11	12	22	6	21	14
	3	13	20	-	18	21	20	22	9	13	19	11
	4	23	15	18	-	7	4	5	25	20	8	6
	5	27	16	21	7	-	6	5	28	21	3	11
	6	21	11	20	4	6	-	2	20	12	10	8
	7	17	12	22	5	5	2	-	23	15	9	10
	8	7	22	9	25	28	20	23	-	12	23	15
	9	5	6	13	20	21	12	15	12	-	12	14
	10	27	21	19	8	3	10	9	23	12	-	10
Customer		20	14	11	6	11	8	10	15	14	10	-

Table B3. Information of AM service providers.

Provider index	Provider type	Basic ordering cost (¥)	Focal service cost (¥)	Opening hours
1	Assistive technology center	0	3,000	Mon–Fri, 09:00–18:00
2	Assistive technology center	0	5,000	Mon–Fri, 09:00–18:00, Sat 09:00–14:00
3	Makerspace	5,000	–	Mon–Fri, 09:00–18:00
4	Makerspace	3,000	–	Mon–Fri, 09:00–18:00
5	Makerspace	5,000	–	Mon–Fri, 09:00–17:00
6	3D printing company	10,000	–	Daily, 09:00–19:00
7	3D printing company	15,000	–	Mon–Fri, 09:00–22:00
8	Individual provider	3,000	–	Daily, 09:00–22:00
9	Individual provider	3,000	–	Daily, 09:00–18:00
10	Individual provider	5,000	–	Daily, 09:00–20:00

Table B4. Information of AM machines.

Provider index	Machine index	Machine Name	Building type	Base area (cm ²)	Setup time (min)	Forming time (hr/cm ³)	Minimum usage time	Basic fee (₩)	Extra fee (₩/hr)
1	1	Shining 3D EP-A650	SLA	3900	45	0.027	2	30,000	8,000
2	2	Stratasys F120	FDM	645	15	0.080	3	15,000	4,000
	3	Stratasys F370	FDM	901.7	15	0.080	3	20,000	6,000
3	4	Cubicon Single Plus	FDM	456	10	0.259	3	5,000	1,500
	5	Cubicon Single Plus	FDM	456	10	0.259	3	5,000	1,500
	6	Cubicon Single Plus	FDM	456	10	0.259	3	5,000	1,500
4	7	Creality Ender-5	FDM	484	10	0.150	4	5,000	1,000
	8	Creality Ender-5	FDM	484	10	0.150	4	5,000	1,000
	9	Creality Ender-5	FDM	484	10	0.150	4	5,000	1,000
	10	Creality Ender-5	FDM	484	10	0.150	4	5,000	1,000
5	11	Cubicon Single Plus	FDM	456	10	0.259	5	4,000	1,000
	12	Cubicon Single Plus	FDM	456	10	0.259	5	4,000	1,000
	13	Cubicon Single Plus	FDM	456	10	0.259	5	4,000	1,000
	14	Cubicon Single Plus	FDM	456	10	0.259	5	4,000	1,000
	15	Stratasys F120	FDM	645	15	0.080	5	5,000	2,000

Continued on next page

Provider index	Machine index	Machine Name	Building type	Base area (cm ²)	Setup time (min)	Forming time (hr/cm ³)	Minimum usage time	Basic fee (¥)	Extra fee (¥/hr)
6	16	Stratasys Fortus 360	FDM	901.7	20	0.103	3	20,000	5,000
	17	Stratasys Fortus 360	FDM	901.7	20	0.103	3	20,000	5,000
	18	Stratasys Fortus 360	FDM	901.7	20	0.103	3	20,000	5,000
	19	3D Systems ProJet 6000	SLA	625	45	0.020	2	50,000	10,000
	20	3D Systems ProJet 6000	SLA	625	45	0.020	2	50,000	10,000
	21	3D Systems ProJet 6000	SLA	625	45	0.020	2	50,000	10,000
7	22	Stratasys F120	FDM	645	15	0.080	3	12,000	2,000
	23	3D Systems Figure 4	DLP	87.6	45	0.027	1	20,000	8,000
	24	3D Systems Figure 4	DLP	87.6	45	0.027	1	20,000	8,000
	25	3D Systems Figure 4	DLP	87.6	45	0.027	1	20,000	8,000
8	26	Creality Ender-5	FDM	484	10	0.150	5	6,000	1,000
	27	Creality Ender-5	FDM	484	10	0.150	5	6,000	1,000
9	28	Cubicon Single Plus	FDM	456	10	0.259	2	1,000	500
	29	Cubicon Single Plus	FDM	456	10	0.259	2	1,000	500
10	30	Creality Ender-5	FDM	484	10	0.150	6	3,000	500